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A Neuro-Fuzzy Approach for Forest Fire Risk Identification Using Aerial Imagery and Meteorological Data

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Abstract

Wildfires are events in which fire spreads uncontrollably, destroying natural ecosystems and causing severe biodiversity loss. Driven by climate change, the frequency and intensity of these events have escalated, making effective risk mitigation a critical global priority. This study proposes a hybrid methodology for the identification and classification of wildfire risk zones, based on the combination of a convolutional neural network (CNN) with U-Net architecture and ResNet-50 backbone for semantic segmentation, together with a fuzzy inference system. The CNN processes high-resolution georeferenced RGB imagery to identify environmental patterns such as vegetation density and combustible organic matter. Evaluated through cross-validation, the CNN achieved a global IoU of 86.73% and a global F1-score of 92.82%, generating a risk classification into low, medium, and high levels. Subsequently, this categorical output is integrated into a fuzzy inference system along with meteorological variables (temperature, humidity, and wind speed). The fuzzy inference system dynamically adjusts the initial risk assessment according to meteorological conditions and generates an updated risk classification for the processed images. This approach significantly improves wildfire risk assessment, providing a data-driven tool for environmental management and early disaster prevention.

Keywords: deep learning; aerial images; semantic segmentation; fuzzy inference system; climatic factors

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1. Introduction

Forest fires, one of the world's most devastating natural disasters, have serious negative effects on the environment, the economy, and society [1]. Globally, it is estimated that fire affects between 3% and 4% of the Earth's land surface annually, being a significant source of greenhouse gas emissions [2]. During the 2024–2025 season, approximately 3.7 million km^2 were burned globally, an area comparable to the territory of India [3]. The growing frequency and severity of these occurrences, exacerbated by human activity and climate change, underscore the critical need for sophisticated instruments to identify and monitor high-risk regions precisely. The spatiotemporal resolution needed for efficient early

warning systems may be lacking in traditional methods of forest fire risk assessment, which frequently rely on static indices or manual interpretation of environmental variables [4]. These approaches often struggle to integrate heterogeneous data sources, thereby systematically limiting their predictive power. Artificial intelligence (AI) techniques have become powerful tools for environmental monitoring in recent years, providing high-accuracy processing of large amounts of multispectral and aerial imagery [5].

Deep learning architectures are particularly well-suited for characterising fire risk because they have demonstrated exceptional performance in tasks such as object detection, semantic segmentation, and land cover classification. Despite these developments, there are still many obstacles to converting AI-based models into functional fire risk assessment systems. The ability of deep learning models to generalise across various geographical locations and environmental conditions, as well as their reliance on sizable, well-labelled datasets, constitute one of their main limitations [6].

Furthermore, purely data-driven approaches often lack interpretability, making it challenging for decision-makers to fully trust or understand the underlying logic of predictions [7]. These limitations emphasise the need for hybrid approaches that combine the advantages of knowledge-based systems and data-driven models to create fire risk assessment frameworks that are more dependable, transparent, and context-aware. By leveraging deep learning techniques through CNNs for semantic segmentation [8], this article seeks to offer a scientific solution. Using this method, researchers can classify and analyse aerial imagery more accurately, allowing them to distinguish between regions with different forest fire risks. A fuzzy logic system is integrated to combine the label map produced by the CNN with relevant meteorological parameters, such as temperature, humidity, and wind speed, because visual information alone cannot fully capture the underlying dynamics of the phenomenon. This integration improves early detection capabilities and provides a more reliable, comprehensible framework by enabling modulation of visually identified risk based on actual environmental conditions. As a result, this greatly advances the management and prevention of forest fire risk.

This paper organises the content as follows. The literature review is presented in Section 2, highlighting various approaches, data sources, and analytical parameters, and summarising pertinent studies on the mapping and identification of forest fire risk areas. The materials and procedures, including data collection, georeferenced RGB image preprocessing, semantic segmentation model implementation, and the design of the suggested hierarchical fuzzy inference system, are covered in detail in Section 3. In Section 4, we present the study's findings, along with graphical outputs that illustrate how we segment fire-risk areas and evaluate the performance of the fuzzy inference system and various semantic segmentation models. In Section 5, we present the primary findings and analyse their implications, constraints, and potential enhancements to our proposed methodology. Finally, we conclude the paper in Section 6 by summarising our contributions and suggesting directions for future research.

2. Literature Review

This section presents a review of techniques for identifying forest fire-prone areas based on the literature. In particular, it examines methods that incorporate data from multispectral and aerial imagery, meteorological parameters, and other pertinent elements, including anthropogenic ones. These approaches typically combine datasets on vegetation traits, meteorological conditions, topographic features, past fire incidents, and human influences. Within this context, deep learning techniques have been widely used to evaluate fire severity and vulnerability, providing valuable information on fire locations and magnitudes. The classification of severity levels reflects the extent of vegetation damage resulting

from wildland fires, which can be classified into levels ranging from very low to very high, depending on the intensity of the impact. Finally, different strategies for modeling the relationship between these factors are examined, highlighting proposals based on artificial intelligence techniques, such as deep learning and fuzzy logic, among others.

Recent studies confirm the usefulness of remote sensing and geospatial analysis for wildfire monitoring. Uttaruk et al. [9] evaluated burned areas in the Omkoi Wildlife Sanctuary using Sentinel-2 images from 2017 to 2020 and spectral indices such as NBR, RBR, and NDWI. Reported burned areas ranged from 0.107 to 1.160 km^2, with overall accuracies between 76.67% and 93.33% and Kappa values from 0.53 to 0.87. Similarly, Laosuwan et al. [10] used Landsat-8 and ΔNBR to map burned areas in Doi Suthep–Pui National Park during 2017–2021, obtaining accuracies of 81.66–86.66% and Kappa coefficients of 0.78–0.84.

These studies demonstrate the reliability of multispectral imagery for burned-area detection, severity evaluation, and post-fire analysis. However, their scope is mainly retrospective, since they focus on areas already affected by fire. In contrast, the present study addresses pre-fire wildfire risk estimation by combining semantic segmentation from aerial imagery with a hierarchical fuzzy inference system that integrates meteorological variables associated with ignition and propagation.

Machine learning has also been widely applied to wildfire risk mapping. Tan and Feng [11] integrated 19 fire-related variables in Hunan Province, China, and compared RF, SVM, and GBDT models; RF obtained the best performance with an AUC of 0.972. Shao et al. [12] used 17,330 VIIRS fire records and multisource environmental data to map fire risk across China. Among RF, SVM, MLP, and GBDT, RF achieved the highest performance, with an AUC of 0.951, accuracy of 87.99%, and F1-score of 88.64%, enabling seasonal municipal-scale risk maps.

Jiang and Peng [13] developed a forest fire prediction framework using meteorological, terrain, anthropogenic, and vegetation variables. Among LR, RF, XGBoost, and SVM, RF again achieved the best results, with 89.53% accuracy, 92.31% recall, 90.28% F1-score, and 0.9555 AUC. SHAP analysis identified minimum relative humidity as the most influential factor. Shao et al. [14] further showed that ensemble learning combining LR, RF, SVM, and GBDT can improve national-scale fire occurrence mapping compared with individual models.

Wijaya and Arymurthy [15] assessed forest and land fire vulnerability in East Nusa Tenggara, Indonesia, using 2022 fire records and fourteen environmental and anthropogenic factors. Seven algorithms were evaluated, and RF obtained the best overall performance, with AUC values of 0.864 for training and 0.742 for testing. Relevant predictors included land cover, NDVI, rainfall, wind speed, slope, and proximity to roads and buildings, confirming the role of vegetation, climate, topography, and human activity in spatial wildfire vulnerability assessment.

Ghali et al. [16] presented a comprehensive review of recent deep learning techniques for forest fire detection, mapping, severity estimation, and spread prediction using satellite remote sensing data, highlighting their potential for wildfire risk analysis, monitoring, and assessment. They present different approaches applied to tasks such as fire detection and mapping, fire severity estimation, and fire spread prediction, demonstrating the potential of these techniques for fire risk analysis, monitoring, and assessment.

Zhang et al. [17] developed a forest fire susceptibility model for Yunnan Province, China, using a CNN. They incorporated twelve spatial predictors including topographic, climatic, anthropogenic, and vegetation variables and converted them into image like formats to allow the CNN to automatically learn spatial patterns associated with fire occurrence. The model achieved high predictive performance (AUC > 0.85), outperforming traditional methods such as logistic regression.

You et al. [18] developed a PSO-CNN model to predict forest fire risk across China using 20 years of data and 11 environmental factors. The particle swarm optimization (PSO) algorithm automatically optimized the CNN architecture, achieving 82.2% accuracy and an AUC of 0.92, outperforming traditional machine learning models.

Zhang, Wang, and Liu [19] compared deep learning models for global wildfire susceptibility. The CNN-2D model, which incorporates spatial neighborhood information, outperformed pixel-based MLP and CNN models, achieving AUC values up to 0.95. Key drivers of wildfire activity were maximum temperature, soil temperature, NDVI, and accumulated precipitation.

Prapas et al. [20] used deep learning to forecast daily wildfire danger from historical Earth observation data, integrating weather, satellite, topographic, and human activity variables. Their ConvLSTM model, which captures spatiotemporal context, achieves the best performance with an AUC of 0.926, outperforming Random Forest. The framework produces high-resolution daily fire danger maps at a national scale.

Le et al. [21] proposed a deep neural computing (Deep-NC) model with three hidden layers and ReLU activation to predict wildfire danger in tropical areas. Using a GIS database with 12 ignition factors (topography, vegetation indices, land use, and climate) and 2530 historical fire records from Vietnam, they tested four optimization algorithms. The Adam-optimized model performed best, achieving an AUC of 0.894 and a Kappa of 0.63, demonstrating its effectiveness for spatial wildfire susceptibility mapping.

Masolele et al. [22] used an attention-enhanced U-Net deep learning model to classify post-deforestation land-use in Ethiopia using Landsat-8, Sentinel-2, and Planet-NICFI imagery. They found that either high spatial resolution (Planet) or high temporal resolution (multi-date Sentinel-2/Landsat-8) is essential for accurate classification, while medium-resolution single-date imagery is insufficient. The attention mechanism improved accuracy for small-scale land-uses.

Sağlam et al. [23] presented a spatio-temporal analysis of forest fire risk and danger using Landsat imagery. The authors develop a methodology based on the integration of remote sensing data and environmental factors, including vegetation characteristics, topography, and proximity to human activities, to generate fire risk and danger indices. These indices are then used to produce risk maps for different time periods, allowing the assessment of changes in fire susceptibility over time. The results demonstrate that satellite imagery provides an effective and cost-efficient tool for mapping and monitoring forest fire risk, achieving high classification accuracy and enabling better support for fire management and planning strategies.

Although CNN-based semantic segmentation models are effective for extracting spatial patterns from aerial imagery, their use as standalone tools for wildfire risk assessment presents important limitations. First, their predictions are primarily based on visual features, such as vegetation density, soil exposure, texture, and land-cover patterns, and therefore they do not explicitly incorporate dynamic meteorological factors that directly influence ignition and propagation processes. Second, purely data-driven models depend strongly on the representativeness of the training data and may show reduced generalization when applied to regions, seasons, or environmental conditions not sufficiently represented in the dataset. Third, CNNs are commonly perceived as black-box models, since the relationship between the learned visual features and the final risk class is not directly expressed through human-interpretable rules.

To address these limitations, this study adopts a neuro-fuzzy hybrid approach. The CNN provides the initial pixel-wise spatial classification, while the fuzzy inference system incorporates expert knowledge through explicit membership functions, linguistic variables, and rule-based reasoning. This structure does not attempt to explain the internal feature

extraction process of the CNN; rather, it provides an interpretable decision layer in which the CNN-derived risk class is adjusted according to meteorological conditions. As a result, the final risk assessment becomes more transparent and traceable, since each modification of the initial CNN output can be associated with specific fuzzy rules involving humidity, temperature, and wind speed. In this sense, the proposed framework combines the pattern-recognition capability of deep learning with the interpretability and contextual reasoning of knowledge-driven fuzzy systems.

The main contributions of this work are as follows: (i) the development and evaluation of semantic segmentation models for the identification of wildfire-prone areas from high-resolution aerial imagery; (ii) the integration of visual information extracted by a CNN with meteorological variables through a hierarchical fuzzy inference system; (iii) the design of a hierarchical fuzzy architecture capable of modeling wildfire ignition and propagation processes while reducing the complexity of the rule base; (iv) the generation of spatially interpretable wildfire risk maps that dynamically adapt to changing environmental conditions; and (v) a comparative evaluation demonstrating the contribution of the fuzzy inference stage with respect to a CNN-only baseline.

3. Materials and Methods

This section describes the methodology adopted in this study, which is systematically structured into six sequential stages as summarized in Figure 1. The proposed framework initiates with (1) Data Acquisition, which involves gathering aerial imagery and relevant meteorological data. This is followed by (2) Data Preparation, dedicated to image preprocessing, patch extraction, manual annotation, and mask generation. In the third stage, (3) Deep Learning Models, the framework handles architecture selection, model training, cross-validation, and performance evaluation to extract robust spatial patterns.

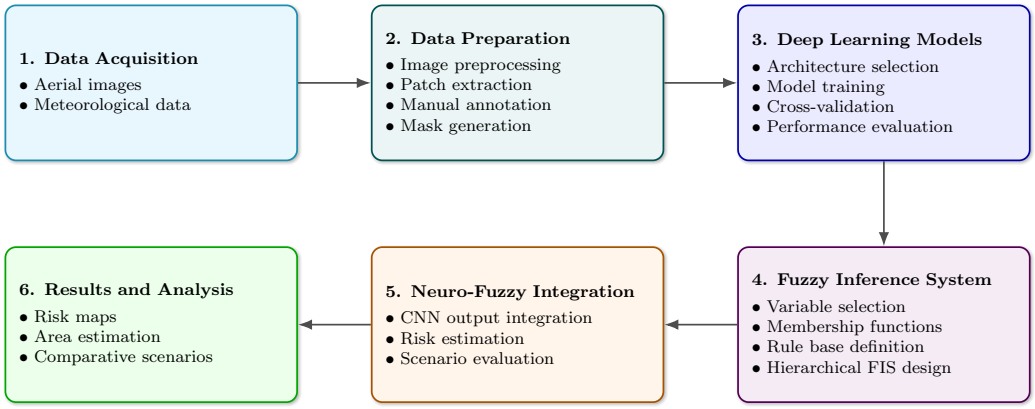


Figure 1. Overall methodological framework

Subsequently, the core analytical processing is executed through a (4) Fuzzy Inference System, which manages variable selection, membership functions, rule base definition, and hierarchical FIS design. These components are then combined in (5) Neuro-Fuzzy Integration to integrate the CNN outputs for risk estimation and scenario evaluation. Finally, (6) Results and Analysis delivers the definitive outputs of the pipeline, including risk maps, area estimation, and comparative scenarios. This structured approach effectively pairs data-driven deep learning with knowledge-based fuzzy logic for enhanced metric quantification and predictive analysis.

3.1. Data Collection and Preprocessing

The image dataset consisted of high-resolution georeferenced RGB imagery exported from the Google Satellite basemap in QGIS. Landsat-8 imagery was used only for prelimi-

nary contextual delimitation of the study area and was not used as raw multispectral input for CNN training. High-resolution images from various environments and geographic locations were selected to ensure a range of conditions in the analysis. The imagery was exported from QGIS at a 1:6000 scale, allowing sufficient spatial detail to identify the main terrain and vegetation features relevant to wildfire susceptibility analysis. Thus, the CNN input corresponds to three-channel RGB images rather than raw multispectral Landsat-8 products. Consequently, preprocessing steps associated with raw multispectral satellite products, such as radiometric calibration, atmospheric correction, spectral-band selection, and cloud masking, were not applied to the final training dataset. Instead, the preprocessing workflow focused on georeferencing verification, visual quality inspection, RGB image export, image–mask alignment, patch extraction, and dataset partitioning. The main image dataset parameters and preprocessing characteristics are summarized in Table 1.

Table 1. Image dataset parameters.

Parameter	Description
Final image source	Google Satellite basemap processed in QGIS
Image type	High-resolution georeferenced RGB imagery
CNN input channels	RGB, three channels
Number of original images	150
Patch size	512 × 512 pixels
Total retained patches	1122
Coordinate handling	Georeferencing verified in QGIS
Associated data	Meteorological records linked to image date and location
Ground-truth generation	Manual semantic annotation
Annotation tool	DigitalSreeni Image Annotator and Toolkit
Risk classes	Low, moderate, and high
Quality control	Image–mask alignment and class-consistency inspection
Dataset DOI	10.5281/zenodo.20669105
Repository structure	Paired Images and Masks folders
File format	PNG image and mask patches
License	Creative Commons Attribution 4.0 International

To improve reproducibility, the image–mask dataset used in this study was deposited in Guillén-Ramos et al. [24]. The repository contains PNG image patches and their corresponding pixel-level masks, organized in paired Images and Masks folders with one-to-one correspondence. The masks encode the initial wildfire susceptibility classes as 0 for low risk, 1 for medium/moderate risk, and 2 for high risk, supporting verification and reproducibility of the CNN-based segmentation models.

The dataset was compiled from different locations across southern Mexico and consisted of 150 high-resolution georeferenced images, each associated with meteorological records corresponding to the date and location of capture. Since the original images had variable spatial dimensions, a patch-based processing strategy was adopted to standardize the CNN input while preserving local spatial details. Each image and its corresponding mask were divided into fixed-size patches of 512 × 512 pixels, see Figure 2. This strategy reduces memory requirements, avoids distortions caused by resizing full images, and is suitable for semantic segmentation tasks involving high-resolution imagery [25].

Let W_{img} and H_{img} denote the width and height of the original image, respectively, and let P denote the fixed patch size. The number of complete patches generated in the horizontal and vertical directions is denoted by N_x and N_y, respectively. The total number of patches extracted from each image is denoted by N_{patches}, as expressed in Equations (1)–(3).

$$N_x = \left\lfloor \frac{W_{\text{img}}}{P} \right\rfloor \quad (1)$$

$$N_y = \left\lfloor \frac{H_{\text{img}}}{P} \right\rfloor \quad (2)$$

$$N_{\text{patches}} = N_x \times N_y \quad (3)$$

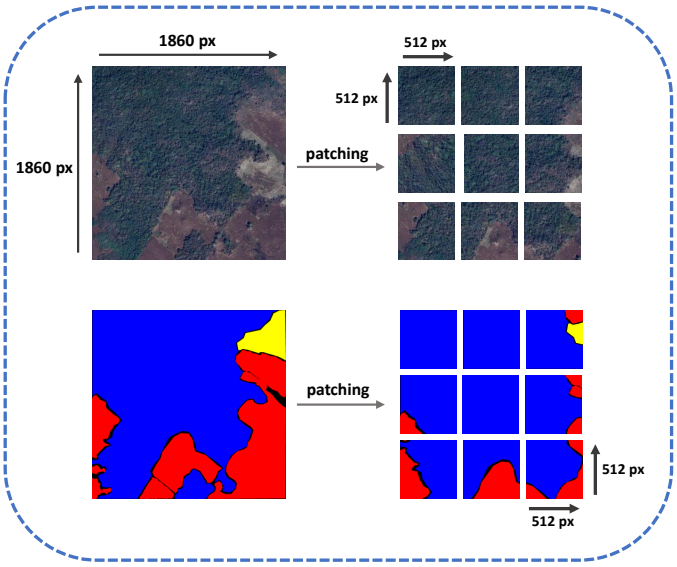


Figure 2. Patching technique applied to images and segmentation masks.

A filtering step was applied to identify and exclude non-informative regions. Patches containing more than 95% background pixels were considered non-useful. However, none of the patches exceeded this threshold; therefore, all 1122 extracted patches were retained. To reduce spatial data leakage, the training and testing split was performed at the original-image level before patch extraction. Specifically, 70% of the original images were assigned to the training set and 30% to the independent testing set. Internal validation during training was performed through a 5-fold cross-validation scheme over the training subset.

To further characterize the annotated dataset, the pixel-level class distribution was computed from all semantic masks. As shown in Table 2, the dataset includes the three wildfire susceptibility classes, with low-risk pixels representing 46.99%, moderate-risk pixels 23.92%, and high-risk pixels 29.08% of the annotated data. This distribution indicates that all classes are represented in the dataset, although low-risk regions are the most frequent class.

Table 2. Pixel-level class distribution of the annotated dataset.

Risk Class	Pixel Count	Percentage
Low	128,963,344	46.99%
Moderate	65,647,323	23.92%
High	79,816,374	29.08%
Total	274,427,041	100.00%

3.2. Ground-Truth Mask Generation

Ground-truth masks were generated through manual semantic annotation of the geo-referenced RGB images using the DigitalSreeni Image Annotator and Toolkit. Since no publicly available pixel-level wildfire risk dataset exists for the study area, the semantic masks were manually constructed to represent three levels of wildfire susceptibility: low,

moderate, and high. These labels were not derived from historical wildfire occurrence inventories; instead, they were defined to describe the intrinsic susceptibility of each land-cover region to wildfire ignition and propagation under favorable environmental conditions.

The annotation criteria were established according to observable environmental characteristics commonly associated with wildfire susceptibility, including vegetation type and density, apparent vegetation condition, fuel continuity, exposed soil, and landscape context. The low-risk class was assigned to areas exhibiting dense and healthy vegetation with high apparent moisture content and limited visual evidence of combustible dry material. The moderate-risk class corresponded to transitional environments characterized by semi-arid vegetation, mixed shrublands, sparse grasslands, or partially dry vegetation, where fuel availability and flammability were considered intermediate. The high-risk class was assigned to regions dominated by exposed soil, dry or degraded vegetation, discontinuous forest cover, and other visual indicators associated with increased fuel flammability.

The annotation process was performed manually by the research team following predefined labeling criteria derived from expert knowledge on wildfire-prone environments. To ensure consistency, all annotated masks were reviewed to verify class coherence across neighboring regions and compliance with the established labeling guidelines. In addition, the original imagery was validated in QGIS to verify georeferencing consistency before patch extraction. Each image–mask pair was subsequently inspected to confirm correct spatial alignment, and an automated filtering procedure was applied to discard patches containing more than 95% background pixels.

The semantic annotation process was carried out by a single annotator following a predefined labeling protocol to ensure consistency throughout the entire dataset. The annotation criteria were established before the labeling process and were uniformly applied to all images, assigning each pixel to one of the three wildfire susceptibility classes according to the visual characteristics described above.

3.3. Semantic Segmentation Convolutional Neural Network Models

Several CNN-based semantic segmentation models were evaluated to identify visual patterns associated with conditions that favor wildfire ignition and spread in high-resolution georeferenced RGB imagery. Through pixel-wise classification, these models extract spatial features related to vegetation, terrain, and other relevant environmental factors.

Due to their demonstrated efficacy in semantic segmentation tasks, a comparative assessment of various U-Net-based architectures was conducted. Specifically, ResNet-based encoder backbones, such as ResNet34 and ResNet50, were evaluated to leverage their feature-extraction capabilities via transfer learning. In parallel, the study examined the performance of a custom, lower-complexity U-Net architecture consisting of a six-layer convolutional structure, see Figure 3. This allowed for a direct performance comparison between established transfer learning models and a lighter, more computationally efficient alternative. We employed a k-fold cross-validation strategy to ensure robust and generalisable results. This method reduces the risk of overfitting and provides a more accurate assessment of performance, particularly when dealing with small datasets [26]. The model can be trained and assessed on various data partitions. The final performance was determined by averaging results across all folds, with each fold used once as a validation set and the remaining folds for training. Several performance metrics commonly used in semantic segmentation tasks were employed to evaluate and select the top-performing model. These include F1-score, Accuracy, Precision, Recall, and Intersection over Union (IoU) [27]. While precision and recall evaluate the model's ability to identify relevant classes accurately and to reduce false positives and false negatives, respectively, accuracy measures the overall percentage of correctly

classified pixels. IoU assesses the overlap between predicted and ground-truth segmentation masks [28], while the F1-Score provides a balanced metric between precision and recall.

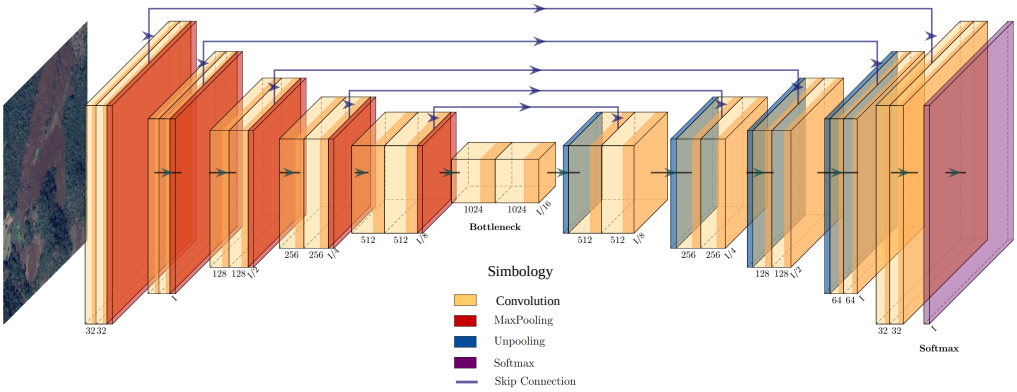


Figure 3. U-Net 6-layer architecture.

3.4. Proposed System

The proposed methodology combines visual and meteorological information through a semantic segmentation CNN and a hierarchical fuzzy inference system (FIS) to estimate forest fire risk.

Initially, the CNN model processes high-resolution georeferenced RGB images and performs pixel-wise classification into three risk categories: low, medium, and high. The softmax output layer generates a label map in which each pixel is assigned to the class with the highest probability, producing a categorical segmentation of the image. This representation captures relevant spatial patterns associated with fire-prone conditions, including vegetation density, dry vegetation, combustible material, and terrain configuration.

The categorical output generated by the CNN is subsequently integrated into the fuzzy inference system as an input variable. To facilitate the fuzzy reasoning process, the predicted classes are represented as singleton discrete numerical values: 0 for low risk, 1 for medium risk, and 2 for high risk. These singleton representations allow the categorical information extracted by the CNN to be directly incorporated into the fuzzy inference process while preserving the semantic interpretation of each risk level. Unlike probability-based approaches, the use of discrete singleton classes enables the FIS to directly adjust the initial risk category according to meteorological conditions. Consequently, regions initially classified as high risk may decrease under favourable meteorological conditions, while areas with medium risk may increase in the presence of adverse environmental factors.

Using expert knowledge from the report by Protección Civil of the state of Chiapas (PROCEDA IN) [29], we selected the meteorological variables employed in the proposed system. This report, together with previous studies [30], defines the relevant meteorological parameters for wildfire risk assessment, including temperature, relative humidity and wind speed, as well as their associated numerical ranges and risk classes. Additionally, the definition of five fire risk levels proposed in previous works [23,31,32] provided a structured basis for the design of the membership functions and fuzzy rule sets.

To model the relative influence of environmental variables more effectively, the proposed FIS adopts a three-level hierarchical structure, as presented in Figure 4, where each level functions as an independent inference subsystem. This design prevents combinatorial explosion in the rule base while improving interpretability and inference efficiency, and it reflects the physical behavior of wildfire processes as well as the relative importance of each variable in ignition and propagation dynamics.

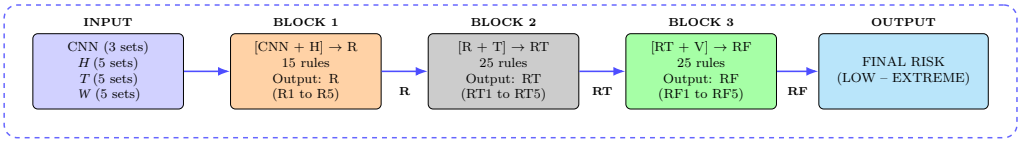


Figure 4. Hierarchical fuzzy inference workflow with multi-level rule blocks for wildfire risk estimation.

Variables such as relative humidity and vegetation-related information extracted by the CNN are prioritised in the first stage because of their strong influence on ignition risk [33]. Temperature is incorporated at an intermediate stage due to its effect on both ignition susceptibility and fire development, whereas wind speed is included in the final stage because of its significant role in fire propagation and intensity. The main environmental factors associated with wildfire ignition and propagation are summarized in Figure 5.

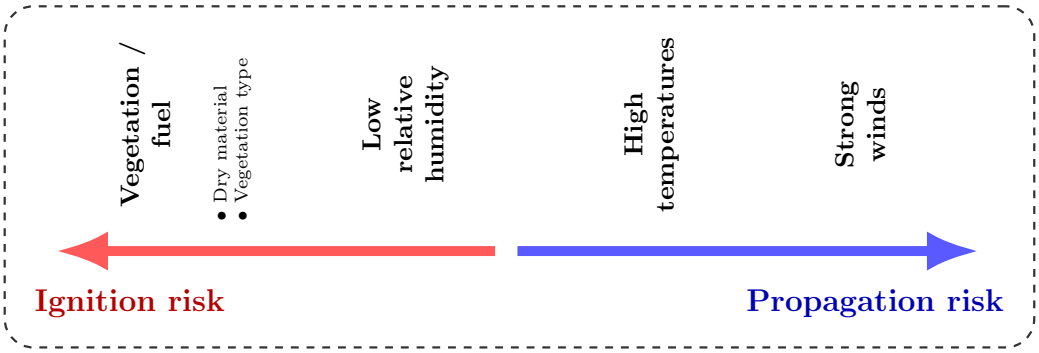


Figure 5. Wildfire risk drivers: Ignition and Propagation.

As a result, the proposed system combines the pattern-recognition capabilities of deep learning with the interpretability and expert-driven reasoning of fuzzy logic, enabling a more coherent and reliable wildfire risk assessment framework.

The overall architecture of the proposed system is illustrated in Figure 6. The model receives as input a georeferenced RGB image represented as a feature tensor of size $[H_{\text{img}} \times W_{\text{img}} \times 3]$ together with meteorological variables, including temperature (T), relative humidity (H) and wind speed (W). These inputs are jointly processed by the CNN and the hierarchical FIS to generate the final forest fire risk estimation.

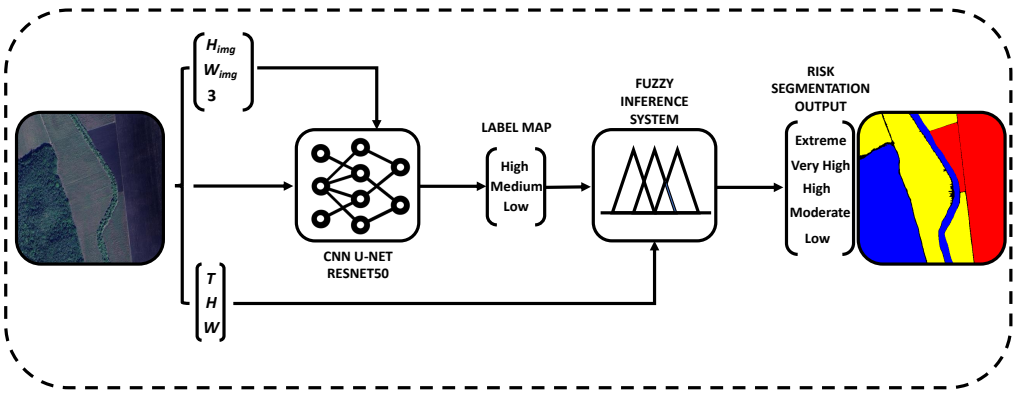


Figure 6. General scheme of the proposed system.

3.5. Three-Level Hierarchical Fuzzy Inference System Configuration

This subsection presents the outcomes of the suggested hierarchical fuzzy inference system (FIS). The analysis focuses on how each inference block behaves, how input variables interact, and the fire risk estimates produced.

Table 3 illustrates the hierarchical structure of the proposed fuzzy inference system, which gradually combines the outputs from the CNN with key meteorological variables. The system consists of three consecutive blocks. In the first block, we generate an intermediate risk indicator R by combining the CNN-based classification with relative humidity. We further refine this output in the second block by integrating temperature, yielding a refined indicator, RT. In the final block, we obtain the overall fire risk level RF by incorporating wind speed. Each block models the uncertainty associated with its respective variable using specific sets of membership functions and operates within predetermined input domains. This hierarchical configuration enhances the overall interpretability and robustness of the risk assessment by enabling a gradual integration of environmental factors.

Table 3. Hierarchical structure of the fuzzy inference system.

Block	Inputs	Domain	MFs	Output	Functional Expression
1	$x_1 = CNN$ $x_2 = Hum$	$CNN \in [0,2]$ $Hum \in [0,100]$	$P_i,\ i = 1...3$ $H_j,\ j = 1...5$	R	$R = f_1(CNN, Humidity)$
2	$x_1 = R$ $x_2 = Temp$	$R \in [0,100]$ $Temp \in [0,45]$	$R_i,\ i = 1...5$ $T_j,\ j = 1...5$	RT	$RT = f_2(R, Temperature)$
3	$x_1 = RT$ $x_2 = Wind$	$RT \in [0,100]$ $Wind \in [0,60]$	$RT_i,\ i = 1...5$ $W_j,\ j = 1...5$	RF	$RF = f_3(RT, Wind)$

The CNN variable represents the class assigned by the neural network for each pixel (0: low, 1: medium, 2: high).

Membership Functions

A key element of the hierarchical fuzzy inference system is the design of membership functions that govern the reasoning process and define how numerical input variables are mapped into linguistic concepts. To ensure consistency with established wildfire risk assessment methodologies, the membership functions employed in this study were defined using expert knowledge derived from the PROCEDA IN framework established in the Programa Especial de Protección Civil del Estado de Chiapas [34]. This framework is intended to support the issuance of wildfire alerts within the state through a color-coded alert mechanism associated with different wildfire risk levels.

According to the PROCEDA IN procedure, wildfire alert conditions are represented using five categories: blue for low risk, green for moderate risk, yellow for high risk, orange for very high risk, and red for extreme risk. In addition to defining the symbolic representation of wildfire danger, the framework establishes numerical threshold intervals associated with each alert level for the main meteorological parameters involved in wildfire occurrence and propagation, including temperature, relative humidity, and wind speed. These operational intervals served as the basis for defining the universe of discourse and the linguistic categories employed in the proposed fuzzy inference system. Consequently, the membership function parameters were designed to preserve consistency with the risk levels and environmental thresholds defined by the PROCEDA IN framework.

Table 4 compiles the membership functions for each input and output variable of the system. The estimated wildfire risk corresponds to the output variable, whereas the input variables include the CNN-derived risk class, temperature, relative humidity, and wind speed. Each variable is represented through a set of linguistic labels such as Low, Moderate, High, Very High, and Extreme, depending on the nature and operational range of the variable.

The universe of discourse of each meteorological variable was established according to environmental ranges commonly associated with wildfire occurrence in the study region and based on the operational thresholds defined within the PROCEDA IN framework. Subsequently, the parameters of the membership functions were adjusted to ensure overlap between adjacent linguistic categories, allowing smooth transitions during the fuzzy reasoning process. Trapezoidal membership functions were employed at the boundary regions to model saturation effects and preserve stability at extreme values, while triangular functions were used for intermediate categories due to their simplicity and interpretability.

Table 4. Membership functions of the fuzzy inference system.

	Risk Level				
Variable	**Low**	**Moderate**	**High**	**Very High**	**Extreme**
CNN	0	1	2	–	–
Temperature	[0, 0, 20, 25]	[22, 26, 30]	[28, 31, 35]	[33, 36, 40]	[38, 41, 45, 45]
Humidity	[55, 65, 100, 100]	[45, 55, 65]	[30, 40, 50]	[15, 25, 35]	[0, 0, 15, 25]
Wind	[0, 0, 15, 20]	[18, 24, 30]	[28, 34, 40]	[38, 44, 50]	[48, 54, 60, 60]
Risk Output	[0, 0, 10, 20]	[15, 30, 40]	[35, 50, 65]	[60, 70, 85]	[80, 90, 100, 100]

The membership function parameters were obtained by transforming the operational reference intervals associated with the PROCEDA IN wildfire alert framework into continuous fuzzy sets. The reference intervals were first used to define the universe of discourse and the linguistic categories for each meteorological variable. Then, boundary categories were represented by trapezoidal functions to model saturation at the minimum and maximum ranges, whereas intermediate categories were represented by triangular functions due to their simplicity and interpretability. For example, the low-temperature category was defined as a left-shoulder trapezoidal function $[0, 0, 20, 25]$, where the interval from 0 to 20 °C represents full membership and the interval from 20 to 25 °C provides a gradual transition toward the moderate category. Similar overlap regions were introduced between adjacent classes for all meteorological variables to avoid abrupt discontinuities in the fuzzy inference process.

Table 5 summarizes the transformation from the reference operational ranges to the final fuzzy membership parameters used in the proposed system. This transformation preserves the five-level alert structure of PROCEDA IN while allowing smooth transitions between linguistic categories during the inference process.

Figure 7 presents the graphical representation of the membership functions associated with the input variables. These plots illustrate the definition and distribution of the linguistic terms used in the fuzzy inference process.

In particular, five linguistic levels were established for temperature, relative humidity, and wind speed, reflecting their influence on wildfire ignition and propagation dynamics. Additionally, the definition of the humidity membership functions reflects the inverse relationship between relative humidity and wildfire risk, where lower humidity values are associated with increased fire susceptibility.

In contrast, the CNN-derived variable corresponds to the categorical output generated by the semantic segmentation model for each pixel. Since this variable represents discrete semantic classes rather than continuous values, singleton membership functions were employed to represent the three risk categories: Low, Moderate, and High. This representation preserves the categorical nature of the CNN classification while enabling its integration within the fuzzy inference framework.

Table 5. Transformation of reference ranges into fuzzy sets.

Variable	Risk Level	Reference Range	Fuzzy Parameters
Temperature	Low	≤20 °C	[0, 0, 20, 25]
Temperature	Moderate	around 26 °C	[22, 26, 30]
Temperature	High	around 31 °C	[28, 31, 35]
Temperature	Very high	around 36 °C	[33, 36, 40]
Temperature	Extreme	≥41 °C	[38, 41, 45, 45]
Humidity	Low risk	≥65%	[55, 65, 100, 100]
Humidity	Moderate	around 55%	[45, 55, 65]
Humidity	High	around 40%	[30, 40, 50]
Humidity	Very high	around 25%	[15, 25, 35]
Humidity	Extreme	≤15%	[0, 0, 15, 25]
Wind speed	Low	≤15 km/h	[0, 0, 15, 20]
Wind speed	Moderate	around 24 km/h	[18, 24, 30]
Wind speed	High	around 34 km/h	[28, 34, 40]
Wind speed	Very high	around 44 km/h	[38, 44, 50]
Wind speed	Extreme	≥54 km/h	[48, 54, 60, 60]
Risk output	Low	0–20	[0, 0, 10, 20]
Risk output	Moderate	20–40	[15, 30, 40]
Risk output	High	40–60	[35, 50, 65]
Risk output	Very high	60–80	[60, 70, 85]
Risk output	Extreme	80–100	[80, 90, 100, 100]

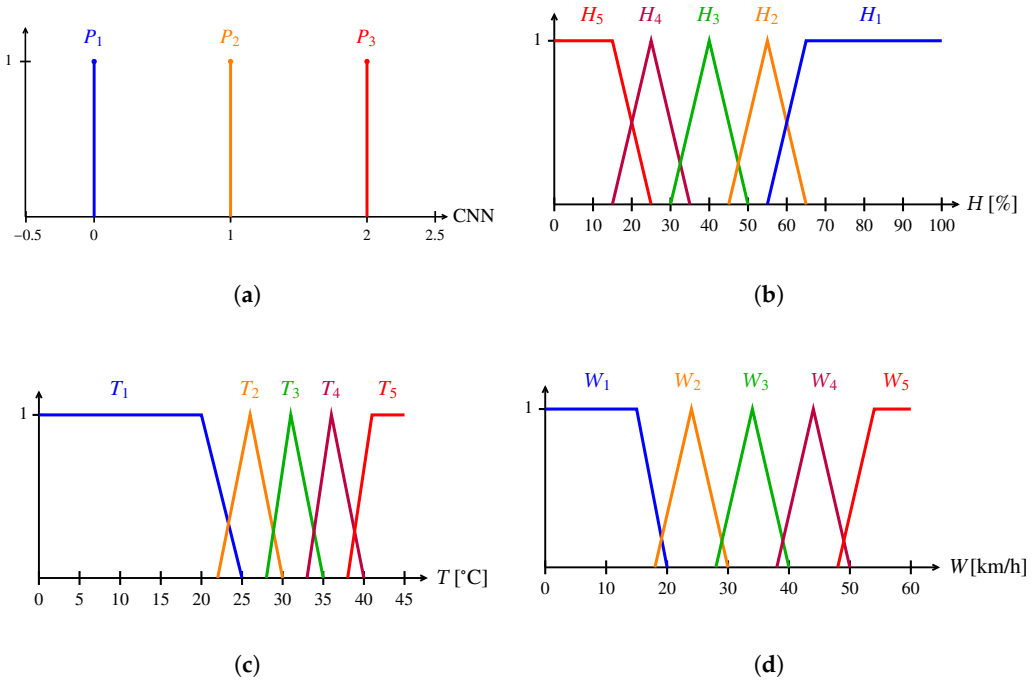


Figure 7. Graphical representation of the input variable membership functions. (**a**) Membership functions for the CNN input variable. (**b**) Membership functions for the Humidity input variable. (**c**) Membership functions for the Temperature input variable. (**d**) Membership functions for the Wind Speed input variable.

Finally, the output variable was defined using five linguistic levels, allowing a more detailed and interpretable representation of the estimated wildfire risk. The fuzzy inference system can successfully integrate heterogeneous data sources and model the nonlinear relationships underlying wildfire risk assessment because the defined membership functions provide a structured and comprehensible mapping between numerical inputs and linguistic concepts.

3.6. Fuzzy Rule Base

The fuzzy rule base defines the core reasoning mechanism of the hierarchical inference system, capturing the relationships between meteorological variables and wildfire risk. The rules were designed to model the physical and environmental dynamics associated with wildfire ignition and propagation processes, considering the interaction between temperature, relative humidity, wind speed, and the initial risk classification provided by the CNN.

The construction of the rule base was supported by expert knowledge and operational criteria reported in wildfire meteorology and fire behavior literature. In particular, the rules were formulated according to the environmental relationships described in the PROCEDA IN framework and studies addressing the influence of meteorological variables on wildfire occurrence and spread. As stated by Almarza from the Instituto Nacional de Meteorología in Meteorología e incendios forestales: El papel de los Servicios Meteorológicos en el apoyo a la prevención de incendios, "meteorology and climate are fundamental elements that largely determine both the probability of wildfire ignition and its subsequent evolution and potential severity." This relationship is especially associated with the effects of high temperatures, low relative humidity, and strong wind conditions on fuel desiccation, fire ignition susceptibility, and flame propagation [35].

Additional studies in wildfire science consistently report that low humidity and elevated temperatures contribute to fuel drying and increase ignition probability, while wind speed directly influences fire spread rate and propagation intensity. Works such as those by Pyne et al. in Introduction to Wildland Fire [36], as well as research in wildfire danger assessment and fire behavior modeling, establish these meteorological variables as key indicators for wildfire risk estimation. Consequently, the proposed fuzzy rules were structured to reproduce these known environmental interactions within the inference system, allowing the model to emulate the gradual and nonlinear behavior observed in real wildfire scenarios. In general, the rule sets describe how variations in temperature, relative humidity, and wind speed interact with the initial CNN-based risk classification to dynamically modulate the final wildfire risk estimation. This approach enables the hierarchical fuzzy inference system to incorporate both visual information extracted from aerial imagery and environmental conditions associated with wildfire development, producing a more interpretable and physically consistent risk assessment.

In the initial stage, the rules emphasize the role of relative humidity and vegetation-related conditions (derived from the CNN), as these factors strongly influence fuel availability and ignition probability. In this regard, Almarza highlights that relative humidity constitutes the most important meteorological variable for explaining daily variations in ignition probability during the period of maximum risk [35]. Consequently, lower humidity levels combined with high-risk visual patterns lead to an increase in the inferred risk, while higher humidity tends to attenuate it. In subsequent stages, temperature contributes to the intensification of fire conditions, acting as an intermediate factor that reinforces or moderates the initial estimation.

The system avoids combinatorial complexity while maintaining interpretability by organising the rules hierarchically, enabling each block to concentrate on particular facets of the fire risk process. Tables 6–8 contain the fuzzy rule sets that correspond to each stage of the hierarchical inference system.

Table 6. Fuzzy rule base of the system considering CNN and Humidity.

#	If	CNN		Humidity	Then	Risk
1	If	P1	and	H1	then	R1
2	If	P1	and	H2	then	R1
3	If	P1	and	H3	then	R2
4	If	P1	and	H4	then	R3
5	If	P1	and	H5	then	R3
6	If	P2	and	H1	then	R1
7	If	P2	and	H2	then	R2
8	If	P2	and	H3	then	R3
9	If	P2	and	H4	then	R4
10	If	P2	and	H5	then	R4
11	If	P3	and	H1	then	R2
12	If	P3	and	H2	then	R3
13	If	P3	and	H3	then	R4
14	If	P3	and	H4	then	R5
15	If	P3	and	H5	then	R5

Table 7. Block 2: Rules for combination of Base Risk (R) with Temperature (T).

#	If	Base Risk (R)		Temperature	Then	Risk T
1	If	R1	and	T1	then	RT1
2	If	R2	and	T1	then	RT1
3	If	R3	and	T1	then	RT2
4	If	R4	and	T1	then	RT3
5	If	R5	and	T1	then	RT3
6	If	R1	and	T2	then	RT1
7	If	R2	and	T2	then	RT2
8	If	R3	and	T2	then	RT3
9	If	R4	and	T2	then	RT4
10	If	R5	and	T2	then	RT4
11	If	R1	and	T3	then	RT2
12	If	R2	and	T3	then	RT3
13	If	R3	and	T3	then	RT4
14	If	R4	and	T3	then	RT5
15	If	R5	and	T3	then	RT5
16	If	R1	and	T4	then	RT2
17	If	R2	and	T4	then	RT3
18	If	R3	and	T4	then	RT4
19	If	R4	and	T4	then	RT5
20	If	R5	and	T4	then	RT5
21	If	R1	and	T5	then	RT3
22	If	R2	and	T5	then	RT4
23	If	R3	and	T5	then	RT5
24	If	R4	and	T5	then	RT5
25	If	R5	and	T5	then	RT5

Table 8. Block 3: Rules for combination of Risk-Temperature (RT) with Wind (W).

#	If	Risk-Temperature (RT)		Wind	Then	Final Risk (RF)
1	If	RT1	and	W1	then	RF1
2	If	RT2	and	W1	then	RF1
3	If	RT3	and	W1	then	RF2
4	If	RT4	and	W1	then	RF3
5	If	RT5	and	W1	then	RF4
6	If	RT1	and	W2	then	RF1
7	If	RT2	and	W2	then	RF2
8	If	RT3	and	W2	then	RF3
9	If	RT4	and	W2	then	RF4
10	If	RT5	and	W2	then	RF5
11	If	RT1	and	W3	then	RF1
12	If	RT2	and	W3	then	RF2
13	If	RT3	and	W3	then	RF3
14	If	RT4	and	W3	then	RF4
15	If	RT5	and	W3	then	RF5
16	If	RT1	and	W4	then	RF1
17	If	RT2	and	W4	then	RF2
18	If	RT3	and	W4	then	RF3
19	If	RT4	and	W4	then	RF4
20	If	RT5	and	W4	then	RF5
21	If	RT1	and	W5	then	RF2
22	If	RT2	and	W5	then	RF3
23	If	RT3	and	W5	then	RF4
24	If	RT4	and	W5	then	RF5
25	If	RT5	and	W5	then	RF5

3.7. Inference Mechanism

The inference mechanism of the proposed fuzzy system follows the Mamdani approach, which consists of four main stages: fuzzification, rule evaluation, aggregation, and defuzzification. As illustrated in Figure 8, this process enables the transformation of crisp inputs into a final quantitative output through a sequence of fuzzy reasoning steps.

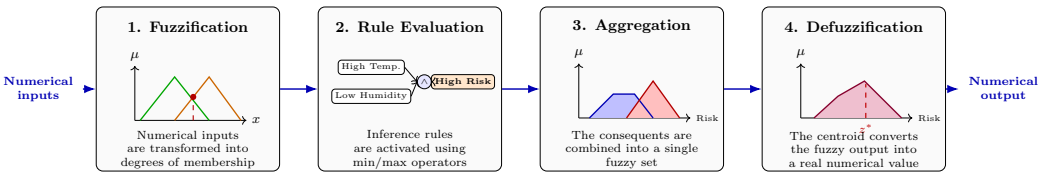


Figure 8. Stages of the Mamdani fuzzy inference system.

A detailed representation of the inference process within the hierarchical structure is illustrated in Figure 9. This figure summarizes the mathematical formulation of each stage, including fuzzification, rule evaluation, aggregation, and defuzzification, for all inference blocks. For the sake of clarity and conciseness, the individual subsections describing each stage focus on the conceptual aspects of the process, while the complete set of expressions governing the system is presented in the figure.

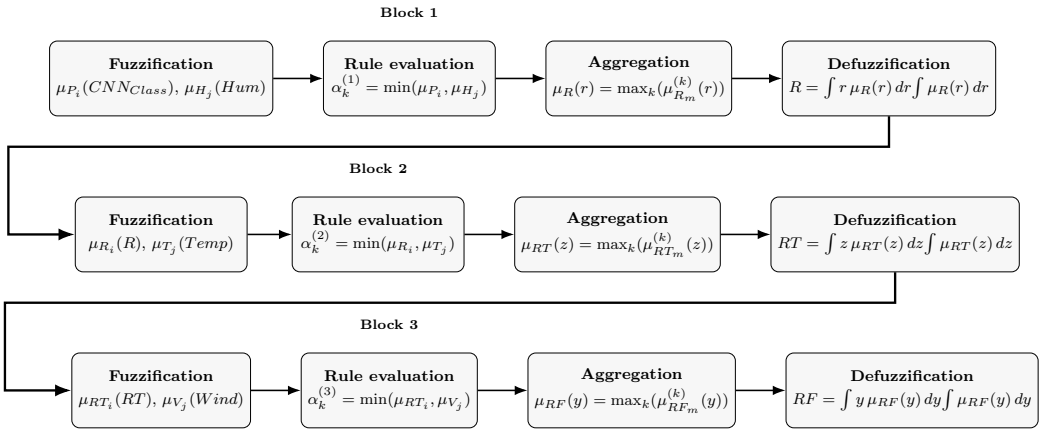


Figure 9. Fuzzy inference process in the proposed hierarchical structure.

3.7.1. Fuzzification

During the fuzzification process, we convert crisp input values into degrees of membership corresponding to predetermined linguistic terms. We can represent numerical values as fuzzy sets by mapping each input variable to its corresponding membership function. The fuzzy inference process that follows is based on this step, enabling the system to manage uncertainty and gradual shifts in risk levels.

3.7.2. Rule Evaluation

Each fuzzy rule's degree of activation, or firing strength, is calculated during the rule evaluation stage using the membership values obtained during the fuzzification procedure. Usually, a set of antecedents joined by logical operators makes up each rule. The minimum operator, which combines the membership degrees of the input variables, is frequently used in the Mamdani approach to model the logical AND. Each rule generates an output fuzzy set that reflects how well the rule conditions are met and scales based on its firing strength. The general rule structure and firing-strength expressions for the three hierarchical blocks are summarized in Table 9.

Table 9. Rule evaluation in the hierarchical fuzzy system.

Block	Input Vars.	General Rule	Firing Strength
1	$(CNN,\ Hum)$	If CNN is P_i and $Humidity$ is H_j then R is R_m	$\alpha_k^{(1)} = \min\left(\mu_{P_i}(CNN), \mu_{H_j}(Humidity)\right)$
2	$(R,\ Temp)$	If R is R_i and $Temp$ is T_j then RT is RT_m	$\alpha_k^{(2)} = \min\left(\mu_{R_i}(R), \mu_{T_j}(Temp)\right)$
3	$(RT,\ Wind)$	If RT is RT_i and $Wind$ is W_j then RF is RF_m	$\alpha_k^{(3)} = \min\left(\mu_{RT_i}(RT), \mu_{W_j}(Wind)\right)$

The rules evaluated by block are described in Tables 6–8.

3.7.3. Aggregation

In the aggregation stage, a single fuzzy set represents each block's overall inference result by combining the outputs of all activated rules. This study employs the maximum operator to integrate the contributions of activated rules, selecting the highest membership degree at each point in the output domain. Using this method, the system selects the most representative answers from the rule base and produces a single fuzzy output for defuzzification. The output fuzzy sets and the aggregation functions used in the three hierarchical blocks are summarized in Table 10.

Table 10. Aggregation process in the hierarchical fuzzy system.

Block	Output per Rule	Aggregation Function	Result
1	$\mu_{R_m}^{(k)}(r) = \min\left(\alpha_k^{(1)}, \mu_{R_m}(r)\right)$	$\mu_R(r) = \max_k\left(\mu_{R_m}^{(k)}(r)\right)$	Fuzzy set $\mu_R(r)$
2	$\mu_{RT_m}^{(k)}(z) = \min\left(\alpha_k^{(2)}, \mu_{RT_m}(z)\right)$	$\mu_{RT}(z) = \max_k\left(\mu_{RT_m}^{(k)}(z)\right)$	Fuzzy set $\mu_{RT}(z)$
3	$\mu_{RF_m}^{(k)}(y) = \min\left(\alpha_k^{(3)}, \mu_{RF_m}(y)\right)$	$\mu_{RF}(y) = \max_k\left(\mu_{RF_m}^{(k)}(y)\right)$	Fuzzy set $\mu_{RF}(y)$

3.7.4. Defuzzification

The defuzzification process converts the aggregated fuzzy sets into quantitative and interpretable outputs. The centroid-based defuzzification expressions used in the three hierarchical blocks are summarized in Table 11. Subsequently, the resulting numerical fire-risk value is mapped into linguistic categories according to the intervals presented in Table 12 [37]. This representation facilitates the interpretation and comparison of different environmental scenarios.

Table 11. Defuzzification process in the hierarchical fuzzy system.

Block	Aggregated Fuzzy Set	Method	Mathematical Expression	Output
1	$\mu_R(r)$	Centroid	$R = \frac{\int r\,\mu_R(r)\,dr}{\int \mu_R(r)\,dr}$	R
2	$\mu_{RT}(z)$	Centroid	$RT = \frac{\int z\,\mu_{RT}(z)\,dz}{\int \mu_{RT}(z)\,dz}$	RT
3	$\mu_{RF}(y)$	Centroid	$RF = \frac{\int y\,\mu_{RF}(y)\,dy}{\int \mu_{RF}(y)\,dy}$	RF

Table 12. Transformation of numerical risk value into linguistic variables (defuzzification).

Risk Value Range	Linguistic Label
$0 \leq RF < 20$	Low (RF1)
$20 \leq RF < 40$	Moderate (RF2)
$40 \leq RF < 60$	High (RF3)
$60 \leq RF < 80$	Very high (RF4)
$80 \leq RF \leq 100$	Extreme (RF5)

The behavior of each stage of the hierarchical fuzzy inference system (FIS) was modeled and analyzed using MATLAB 2025b, allowing the evaluation of the fuzzy reasoning process, the interaction between membership functions, and the influence of meteorological variables on wildfire risk estimation. For each inference stage, a three-dimensional response surface was generated to represent the relationship between the input variables and the corresponding output of the block. These surfaces provide a visual and quantitative interpretation of how the fuzzy rules and membership functions interact to produce the output. As shown in Figure 10, each surface illustrates the nonlinear mapping between the input variables and the inferred risk level, highlighting the influence and relative contribution of each parameter within the hierarchical structure. This analysis allows for a better understanding of the system's behavior and facilitates the validation of the designed rule base and membership functions.

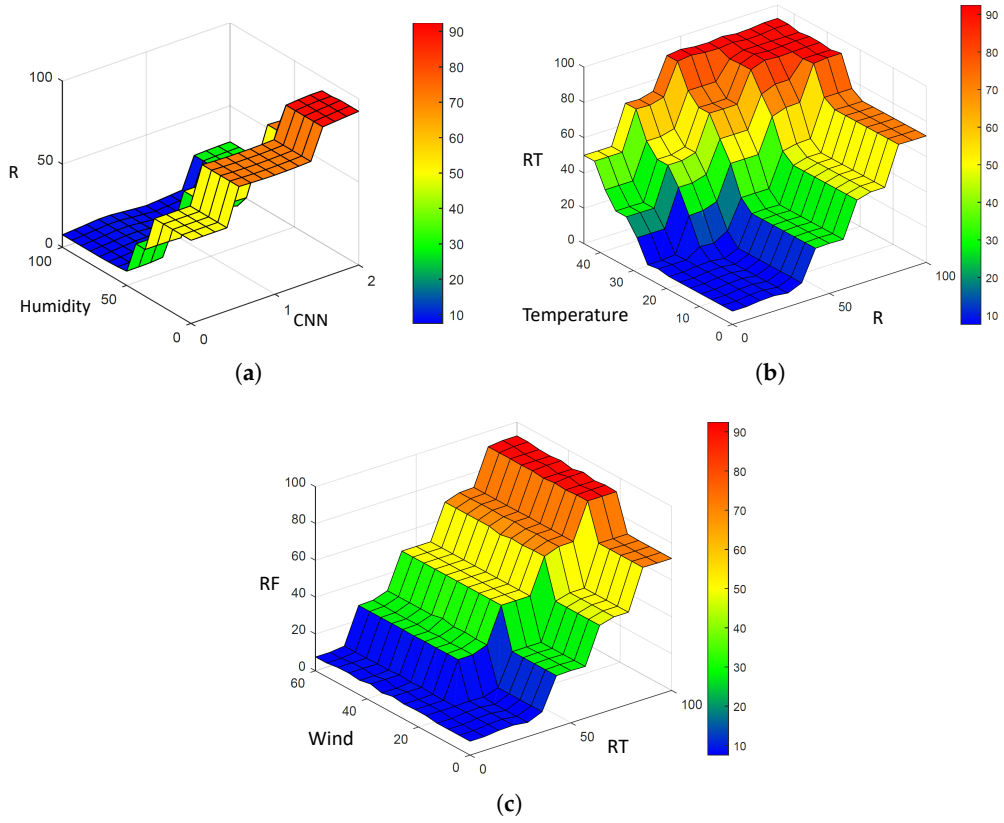


Figure 10. 3D surfaces of the hierarchical fuzzy inference system (FIS) blocks. (**a**) Surface of the first block, relating CNN output (cnn_class) and humidity to obtain the intermediate variable *R*. (**b**) Surface of the second block, relating *R* and temperature to obtain the intermediate variable *RT*. (**c**) Surface of the third block, relating *RT* and wind speed to obtain the final output variable RF.

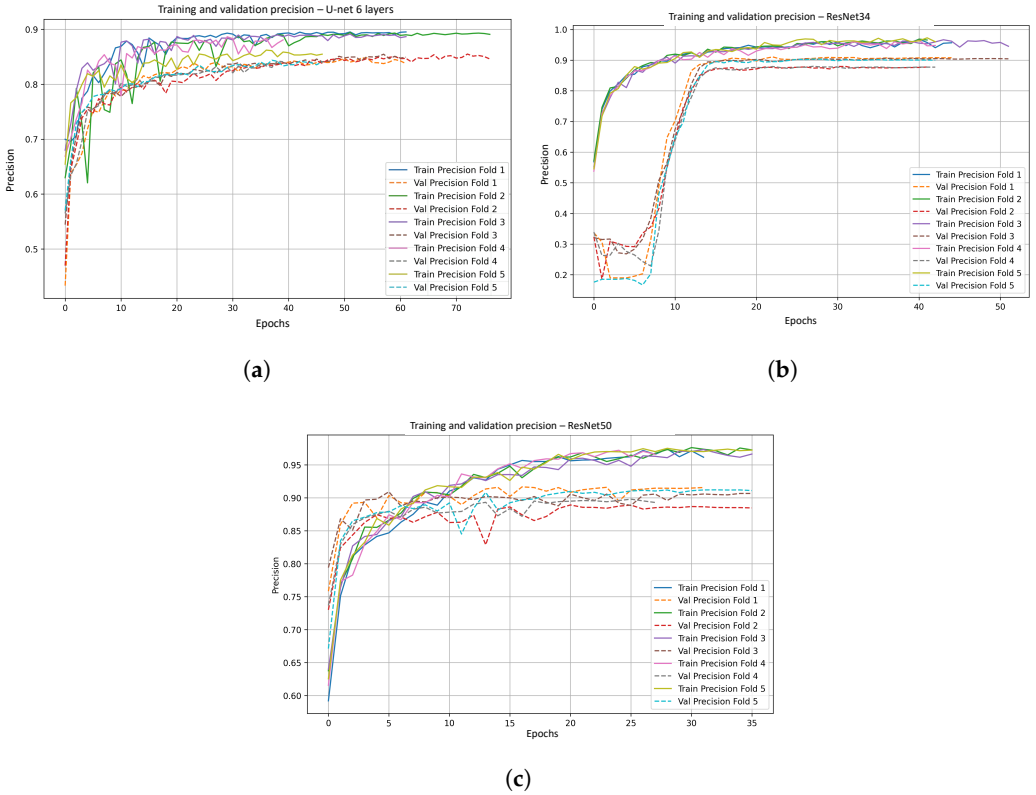
4. Results

The performance of the hierarchical fuzzy inference system (FIS) and the semantic segmentation models are among the outcomes of the suggested methodology presented in this section. To determine the best architecture based on quantitative metrics, a comparative analysis of segmentation models is first presented. The proposed FIS delivers results demonstrating its effectiveness in integrating meteorological and visual data to estimate forest fire risk.

4.1. Comparative Evaluation of Semantic Segmentation Models

The best architecture for the suggested application was determined by training and assessing the semantic segmentation models. To ensure efficient computation and shorter training times, all models were implemented and trained on the Google Colab platform, which used an NVIDIA A100 GPU. We used a 5-fold cross-validation strategy ($k = 5$) to provide a reliable and objective assessment. This method improves the evaluation of generalisation capability and lowers the risk of overfitting by enabling each model to be trained and validated across various data partitions. Standard metrics for semantic segmentation, such as Accuracy, Precision, Recall, F1-Score, and Intersection over Union (IoU), were used to assess the model performance for each fold.

Figure 11 shows the training and validation precision curves, highlighting the models' learning stability and convergence behavior. The close alignment between training and validation performance demonstrated that the evaluated architectures generally converged consistently with no discernible overfitting.



(**a**) (**b**)

(**c**)

Figure 11. Comparative results of the U-Net architectures across folds. (**a**) Training and validation results for each fold using the 6-layer U-Net architecture. (**b**) Training and validation results for each fold using U-Net with ResNet34 backbone. (**c**) Training and validation results for each fold using U-Net with ResNet50 backbone.

Tables 13–15 present the pixel-level quantitative results obtained through 5-fold cross-validation for the evaluated semantic segmentation architectures. For each fold, Accuracy, macro Precision, macro Recall, macro F1-score, and macro IoU are reported, together with their mean and standard deviation across folds. In addition, each table includes a GLOBAL row, computed by aggregating all validation pixels from the five folds and evaluating the metrics over the resulting combined set. Accordingly, the mean values characterize the stability of each model across different validation partitions, whereas the GLOBAL values provide an overall estimate of the segmentation performance.

Table 13. Metrics for the 6-layer U-Net.

Fold	Accuracy	Precision	Recall	F1-Score	IoU
Fold 1	0.8951	0.8919	0.8873	0.8884	0.8020
Fold 2	0.8926	0.8887	0.8863	0.8853	0.7961
Fold 3	0.8913	0.8865	0.8839	0.8846	0.7965
Fold 4	0.8864	0.8832	0.8756	0.8778	0.7855
Fold 5	0.8621	0.8566	0.8557	0.8531	0.7466
Mean	0.8855	0.8814	0.8778	0.8778	0.7853
Std	0.0120	0.0127	0.0118	0.0128	0.0201
GLOBAL	0.8855	0.8810	0.8777	0.8778	0.7849

Table 14. Metrics for U-Net ResNet-34.

Fold	Accuracy	Precision	Recall	F1-Score	IoU
Fold 1	0.9247	0.9089	0.9173	0.9091	0.8409
Fold 2	0.9219	0.8792	0.8766	0.8746	0.8012
Fold 3	0.9279	0.9067	0.9130	0.9076	0.8421
Fold 4	0.9251	0.8799	0.8824	0.8789	0.8100
Fold 5	0.9285	0.9034	0.9135	0.9059	0.8369
Mean	0.9256	0.8956	0.9006	0.8952	0.8262
Std	0.0027	0.0148	0.0194	0.0170	0.0192
GLOBAL	0.9204	0.9212	0.9204	0.9202	0.8533

Table 15. Metrics for U-Net ResNet-50.

Fold	Accuracy	Precision	Recall	F1-Score	IoU
Fold 1	0.9331	0.9165	0.9279	0.9189	0.8565
Fold 2	0.9345	0.8891	0.8938	0.8875	0.8228
Fold 3	0.9285	0.9056	0.9116	0.9061	0.8403
Fold 4	0.9327	0.8903	0.8881	0.8868	0.8230
Fold 5	0.9357	0.9097	0.9212	0.9125	0.8480
Mean	0.9329	0.9022	0.9085	0.9024	0.8381
Std	0.0024	0.0108	0.0154	0.0131	0.0134
GLOBAL	0.9329	0.9281	0.9284	0.9282	0.8673

Among the evaluated configurations, the U-Net model with ResNet-50 backbone achieved the best overall performance. This architecture obtained the highest mean Accuracy (0.9329), mean F1-score (0.9024), and mean IoU (0.8381) across folds. Moreover, its GLOBAL metrics reached an F1-score of 0.9282 and an IoU of 0.8673, which are the values reported in the Abstract. Compared with the 6-layer U-Net, the ResNet-50 backbone improved the mean IoU from 0.7853 to 0.8381 and the GLOBAL IoU from 0.7849 to 0.8673. Based on these results, U-Net ResNet-50 was selected as the semantic segmentation model for integration with the hierarchical fuzzy inference system.

The differences among the evaluated architectures are mainly associated with the use of transfer learning. Unlike the 6-layer U-Net, which was trained from scratch, the U-Net ResNet-34 and U-Net ResNet-50 models use ImageNet-pretrained encoder backbones. These feature extractors provide more robust visual representations, improving the learning process when working with datasets of moderate size. As shown in Table 16, the ResNet-based models outperformed the conventional U-Net in all evaluation metrics. The U-Net ResNet-50 achieved the best results, with an IoU of 0.8381 and an Accuracy of 0.9329, representing absolute gains of 5.28 and 4.74 percentage points, respectively, over the 6-layer U-Net. Improvements were also observed in Precision, Recall, and F1-Score.

Table 16. General metrics of evaluated models and absolute improvement percentage.

	Evaluated Architectures			
Evaluation Metric	**U-Net 6 Layers**	**U-Net ResNet-34**	**U-Net ResNet-50**	**Net Absolute Improvement (ResNet-50 vs. 6 Layers)**
Accuracy	0.8855	0.9256	0.9329	+4.74%
Precision	0.8814	0.8956	0.9022	+2.08%
Recall	0.8778	0.9006	0.9085	+3.07%
F1-Score	0.8778	0.8952	0.9024	+2.46%
IoU	0.7853	0.8262	0.8381	+5.28%

From a practical perspective, the 6-layer U-Net requires less computational effort but has a lower capacity to extract complex spatial features. In contrast, the ResNet-based models provide deeper representations and better generalization. Since U-Net ResNet-50 achieved the highest segmentation performance, it was selected for integration with the hierarchical fuzzy inference system.

4.2. Comparison with Mainstream Semantic Segmentation Models

To further assess the performance of the CNN segmentation module used in the proposed neuro-fuzzy framework, a comparison was conducted against three recent and representative semantic segmentation models: Mask2Former, OneFormer, and Segment Anything. Mask2Former uses a masked-attention transformer architecture for universal image segmentation [38]. OneFormer introduces a task-conditioned transformer framework trained for multiple segmentation tasks within a unified architecture [39]. Segment Anything is a promptable foundation model designed for general-purpose image segmentation [40].

All models were trained and evaluated using the same dataset, input patch size, RGB channels, class labels, data partitioning, and cross-validation protocol adopted for the segmentation experiments. The comparison was performed using Accuracy, Precision, Recall, F1-score, and IoU. Table 17 summarizes the quantitative comparison. The U-Net ResNet-50 model achieved the highest performance in all metrics, with an Accuracy of 0.9329, Precision of 0.9022, Recall of 0.9085, F1-score of 0.9024, and IoU of 0.8381. Among the compared models, Mask2Former and OneFormer showed competitive performance due to their transformer-based segmentation design, while Segment Anything also achieved strong results but remained slightly below the proposed model. Overall, these results indicate that U-Net ResNet-50 provides a more consistent balance between pixel-level classification accuracy and spatial segmentation quality, supporting its selection as the segmentation module of the proposed neuro-fuzzy framework.

Table 17. Comparison with recent mainstream semantic segmentation models.

Model	Accuracy	Precision	Recall	F1-Score	IoU
U-Net ResNet-50	0.9329	0.9022	0.9085	0.9024	0.8381
Mask2Former	0.9298	0.8997	0.9056	0.8998	0.8349
OneFormer	0.9286	0.8988	0.9041	0.8987	0.8328
Segment Anything	0.9264	0.8969	0.9017	0.8966	0.8295

4.3. Experimental Results and Case Studies Analysis

To demonstrate the operation of the proposed system for the identification and classification of wildfire risk areas, a representative study region was selected considering the environmental and geographical conditions of the area. Figure 12 presents the high-resolution aerial image selected for the experiments. Through a patching process using sub-images of 640 × 640 pixels, one representative region was selected for the evaluation of the proposed methodology, highlighted in red within the study area. The geographical characteristics associated with the complete study image are summarized in Table 18; these parameters are used throughout the experiments presented in the following sections.



Figure 12. Delimited study area used for the experimental evaluation of the proposed neuro-fuzzy system. The red region corresponds to the selected image patch employed in the case studies.

Table 18. Characteristics of the study area.

Parameter	Value
Coordinate reference system	EPSG:4326
Latitude	North: 17.7072406 South: 17.6949241
Longitude	West: −97.2391421 East: −97.2157585
Image resolution	2358 × 1242 pixels
Selected patch size	640 × 640 pixels
Image file size	5.2 MB
Scale	1:6000
Date	May 2025

The analyzed image corresponds to a section of the Mixteca Alta in the state of Oaxaca, a region located in southern Mexico that historically ranks as one of the states with the highest annual incidence of wildfires. The landscape environment in this zone is characterized by a complex topography of mountainous terrains and high intermontane valleys, where patches of xerophytic scrub vegetation, agricultural areas, and altered pine-oak forests coexist. Due to the cyclical drought that severely affects the region during the dry season, the vegetative cover accumulates high water stress, rendering this territory highly vulnerable and an ideal scenario for environmental research and wildfire risk modeling.

To evaluate the behavior of the proposed neuro-fuzzy system, a set of experiments was conducted using the selected aerial image under different meteorological scenarios. In each meteorological scenario, the environmental parameters, including temperature, relative humidity, and wind speed, were modified in order to analyze how these variables influence and modulate the estimated wildfire risk. A total of four experimental scenarios are presented, each corresponding to a specific combination of meteorological conditions.

For each scenario, a table summarizes the defined input parameters together with the numerical outputs obtained at each stage of the hierarchical fuzzy inference system. Additionally, the segmentation results are visually presented, including the original aerial image, the CNN-based risk classification, and the outputs generated by each inference block of the hierarchical FIS. This representation allows a detailed observation of how the wildfire risk evolves and is progressively adjusted throughout the fuzzy inference process

according to the environmental conditions. A color scale is also incorporated to represent the different risk levels, ranging from low to extreme, facilitating the interpretation of the spatial distribution of the estimated wildfire risk. Overall, these experiments demonstrate the capability of the proposed system to dynamically adapt risk estimates according to varying meteorological conditions.

Although the fuzzy membership functions and meteorological thresholds were defined according to the PROCEDA IN wildfire risk framework developed by the Civil Protection Agency of the neighboring state of Chiapas, their application to the study area is supported by the environmental and climatic similarities shared by both regions. Chiapas and Oaxaca belong to southern Mexico and exhibit comparable seasonal climatic patterns, wildfire regimes, and vegetation types associated with fire-prone ecosystems. Moreover, the PROCEDA IN framework establishes expert-defined relationships between temperature, relative humidity, wind speed, and wildfire risk that describe the physical influence of these variables on wildfire ignition and propagation rather than location-specific characteristics. Consequently, these expert criteria provide a suitable knowledge base for the proposed hierarchical fuzzy inference system. Nevertheless, because the framework is modular, the membership functions and fuzzy rules can be recalibrated using regional technical guidelines or local expert knowledge when applying the proposed methodology to other geographic areas with different environmental conditions.

4.3.1. Meteorological Scenario 1

Scenario 1 evaluates the system's response under transitional environmental conditions in the Mixteca Alta region, characterized by intense heat paired with a moderate presence of atmospheric moisture. As specified in Table 19, the input parameters consist of an elevated temperature of 38 °C, a relative humidity of 40%, and a wind speed of 28 km/h. The corresponding spatial distribution of these risk segmentation results is illustrated in Figure 13.

Table 19. Risk assessment results for meteorological scenario 1.

Conditions	**Humidity = 40%, Temperature = 38 °C, Wind = 28 km/h**		
CNN	**Block 1**	**Block 2**	**Block 3**
0	28.19 (Moderate)	50 (High)	50 (High)
1	61.34 (High)	78.09 (Very high)	72.13 (Very high)
2	71.67 (Very high)	91.44 (Extreme)	91.07 (Extreme)

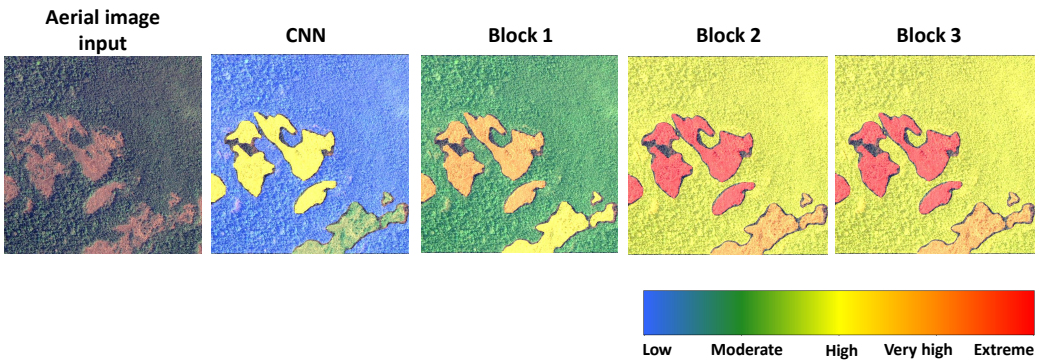


Figure 13. Risk segmentation results for Scenario 1.

The hierarchical flow of the FIS demonstrates a progressive and balanced risk modulation. In Block 1, combining the CNN categorical outputs with the 40% relative humidity allows the system to mitigate the upward trend; this moisture percentage acts as a buffer that stabilizes risk progression compared to absolute drought conditions, placing low-susceptibility zones at 28.19 (Moderate). In Block 2, the integration of this intermediate

result with the high temperature of 38 °C accounts for the environmental thermal load, causing highly vulnerable areas to reach 'Extreme' levels (91.44). Upon reaching Block 3, the introduction of the 28 km/h moderate wind speed determines the final risk output, keeping maximum indicators stable (91.07) without reaching the critical saturation seen in drier scenarios, which proves the sensitivity of the FIS to early moisture-buffering factors.

4.3.2. Meteorological Scenario 2

Scenario 2 evaluates the system's response under safe or low-criticality atmospheric conditions, which are characteristic of periods with high moisture content or the onset of the rainy season in the region. As detailed in Table 20, the input parameters consist of a moderate temperature of 26 °C, a high relative humidity of 60%, and a light wind speed of only 11 km/h. The mapping of these risk segmentation results is displayed in Figure 14.

Table 20. Risk assessment results for meteorological scenario 2.

Conditions	**Humidity = 60%, Temperature = 26 °C, Wind = 11 km/h**		
CNN	**Block 1**	**Block 2**	**Block 3**
0	8.56 (Low)	7.52 (Low)	7.52 (Low)
1	18.53 (Low)	21.35 (Moderate)	8.72 (Low)
2	40.08 (High)	50 (High)	28.33 (Moderate)

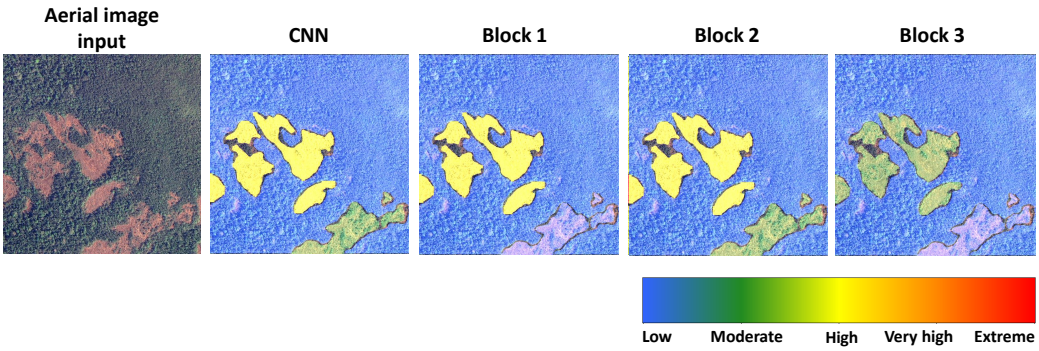


Figure 14. Risk segmentation results for Scenario 2.

The hierarchical flow of the FIS clearly demonstrates a risk mitigation and downward stabilization trend. In Block 1, processing the CNN spatial output alongside the high 60% relative humidity acts as an immediate and powerful environmental buffer, keeping the indicators well-controlled and classifying lower susceptibility zones as 'Low' (8.56 and 18.53). Subsequently, in Block 2, the introduction of the mild 26 °C temperature maintains the risk values within safe margins, capping the highest susceptibility area at 50 (High). Finally, upon reaching Block 3, the incorporation of the light 11 km/h wind speed establishes the final risk output by actively dampening high threat levels: intermediate zones drop to 8.72 (Low), while the most vulnerable areas initially mapped by the CNN are substantially mitigated to 28.33, categorized only as 'Moderate'. These outcomes validate the capability of the hierarchical FIS to dynamically downscale risk estimates when meteorological conditions reduce the inferred wildfire risk right from the initial moisture stage.

4.3.3. Meteorological Scenario 3

Scenario 3 evaluates the critical atmospheric conditions characteristic of the end of the dry season (typically from March to May). During this period, the regional vegetation and organic fuel have accumulated severe hydric stress due to cyclical drought, increasing ignition susceptibility. As detailed in Table 21, this scenario simulates extreme meteorological conditions: an elevated temperature of 40 °C, a critically low relative humidity of

13%, and strong wind speeds of 45 km/h. The spatial representation of the resulting risk segmentation is presented in Figure 15.

Table 21. Risk assessment results for meteorological scenario 3.

Conditions	**Humidity = 13%, Temperature = 40 °C, Wind = 45 km/h**		
CNN	**Block 1**	**Block 2**	**Block 3**
0	50 (High)	71.81 (Very high)	71.71 (Very high)
1	71.67 (Very high)	91.82 (Extreme)	92.16 (Extreme)
2	92.48 (Extreme)	91.82 (Extreme)	92.16 (Extreme)

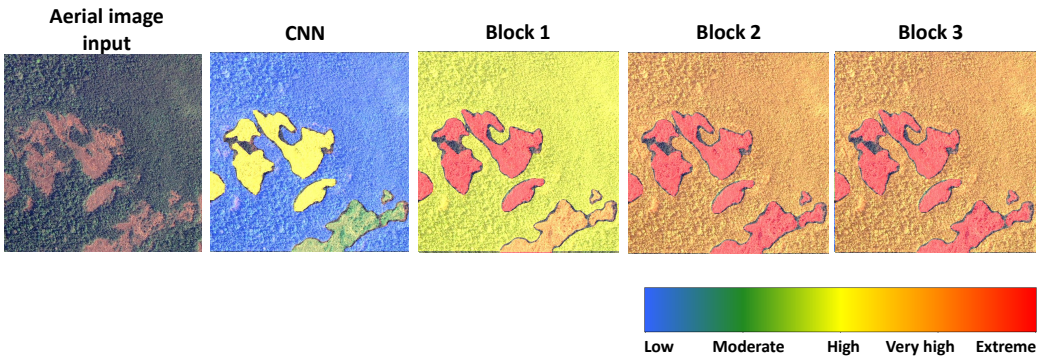


Figure 15. Risk segmentation results for Scenario 3.

Through the structure of the hierarchical fuzzy inference system (FIS), it is possible to observe how the risk levels are sequentially modulated. In Block 1, combining the CNN spatial classification with the critical 13% humidity deficit forces an immediate rise in the baseline risk indicators due to the extreme dryness of the fuel. Subsequently, in Block 2, integrating the output of Block 1 with the extreme 40 °C temperature sharply escalates the threat toward 'Extreme' levels (91.82) for the most vulnerable classes. Finally, in Block 3, the incorporation of the 45 km/h wind speed acts as the definitive propagation catalyst, consolidating the final risk output at critical values of up to 92.16 (Extreme). These outcomes highlight the capability of the FIS to chain environmental variables and accurately simulate high-danger scenarios driven by dryness, thermal load, and severe ventilation.

4.3.4. Meteorological Scenario 4

Scenario 4 evaluates the system's response under conflicting atmospheric conditions, which are characteristic of dry winter fronts or seasonal transitional days in the region. As detailed in Table 22, the input parameters consist of a low temperature of 19 °C, a low relative humidity of 27%, and a wind speed of 33 km/h. The corresponding mapping of the risk segmentation results is illustrated in Figure 16.

Table 22. Risk assessment results for meteorological scenario 4.

Conditions	**Humidity = 27%, Temperature = 19 °C, Wind = 33 km/h**		
CNN	**Block 1**	**Block 2**	**Block 3**
0	50 (High)	28.33 (Moderate)	28.29 (Moderate)
1	71.72 (Very high)	50 (High)	50 (High)
2	92.09 (Extreme)	71.67 (Very high)	71.71 (Very high)

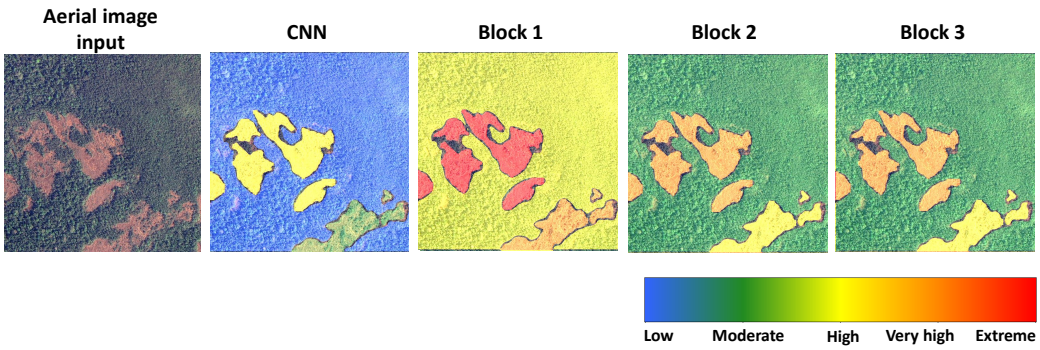


Figure 16. Risk segmentation results for Scenario 4.

The sequential behavior of the hierarchical FIS highlights how these opposing environmental factors interact with the initial CNN spatial classification. In Block 1, evaluating the CNN output alongside the 27% humidity deficit immediately detects environmental dryness, forcing an upward adjustment of the baseline risk into values of 50 (High) and 71.72 (Very High). However, upon moving to Block 2, integrating this dry state with the cool 19 °C temperature acts as a thermal brake that counteracts exponential growth, stabilizing and redistributing the risk values into 'Moderate' (28.33) and 'Very High' (71.67) categories. Finally, in Block 3, the incorporation of the 33 km/h wind speed establishes the final risk output, keeping the indicators stable at 'Moderate' (28.29) and 'Very High' (71.71) levels. This scenario demonstrates how the temperature block effectively restrains the alert from escalating into 'Extreme' categories due to the low thermal load, despite the presence of significant wind and low moisture.

4.4. Spatial Surface Estimation

In addition to the qualitative analysis of the segmentation results, a spatial estimation process was performed to determine the real-world surface represented by the detected wildfire risk regions. Since the study area is defined in the EPSG:4326 geographic coordinate system, geodesic calculations were applied to estimate the physical dimensions of the analyzed region.

The angular latitude and longitude differences of the study area were first calculated as:

$$\Delta\phi = |\mathrm{Lat_N} - \mathrm{Lat_S}| \tag{4}$$

$$\Delta\lambda = |\mathrm{Lon_E} - \mathrm{Lon_W}| \tag{5}$$

The linear distances corresponding to latitude and longitude were then estimated using:

$$D_{\mathrm{lat}} \approx \Delta\phi \cdot 111.13 \tag{6}$$

$$D_{\mathrm{lon}} \approx \Delta\lambda \cdot 111.32 \cdot \cos(\phi) \tag{7}$$

where ϕ corresponds to the mean latitude of the study region. The total surface area was approximated as:

$$A = D_{\mathrm{lat}} \cdot D_{\mathrm{lon}} \tag{8}$$

Using the geographical limits of the selected region, the analyzed area was estimated to cover approximately 3.39 km^2.

Subsequently, the spatial resolution of the aerial image was estimated by relating the physical dimensions of the study area with the image dimensions in pixels. The ground sampling distance (GSD) for each axis was calculated as:

$$\text{Res}_x = \frac{D_{\text{lon}}}{N_x} \tag{9}$$

$$\text{Res}_y = \frac{D_{\text{lat}}}{N_y} \tag{10}$$

where N_x and N_y correspond to the horizontal and vertical image dimensions in pixels. The estimated spatial resolutions were approximately 1.05 m/px and 1.10 m/px for the horizontal and vertical directions, respectively.

The surface represented by each pixel was then estimated as:

$$A_{\text{px}} = \text{Res}_x \cdot \text{Res}_y \tag{11}$$

resulting in an approximate area of 1.15 m^2/px.

Finally, the segmented masks generated by the proposed neuro-fuzzy system were analyzed to estimate the physical coverage associated with each wildfire risk category. The total area occupied by each class was computed as:

$$A_{\text{risk}} = N_{\text{px}} \cdot A_{\text{px}} \tag{12}$$

where N_{px} represents the number of pixels classified within a specific wildfire risk category.

To quantify the wildfire risk maps, the spatial distribution of risk classes was computed for each meteorological scenario. The analysis included the number of classified pixels, spatial coverage, and estimated physical area for each risk category. These results allow a direct comparison of how the hierarchical FIS modifies the spatial extent of wildfire risk under different temperature, relative humidity, and wind speed conditions. The total detected area corresponds to the informative region classified by the model, whereas the remaining percentage is associated with background or non-analyzed pixels. The results are summarized in Table 23.

These calculations provide a physical interpretation of the segmentation results, transforming pixel-level detections into measurable spatial units useful for wildfire monitoring and environmental risk assessment.

4.5. Wildfire Risk Modulation Under Representative Meteorological Scenarios

Figure 17 shows the risk modulation process for four representative meteorological scenarios associated with the seasonal conditions of the study region. Each scenario uses average temperature, relative humidity, and wind speed values for the corresponding months, as indicated in the figure. The initial susceptibility level generated by the CNN is sequentially adjusted by the hierarchical FIS, allowing the influence of each meteorological variable on the final wildfire risk estimation to be interpreted.

As shown in Figure 17, the CNN provides an initial susceptibility level based on visual features extracted from the aerial image. This value is then adjusted by relative humidity in Block 1, temperature in Block 2, and wind speed in Block 3, producing the final wildfire risk level. The trajectories indicate that the proposed hierarchical FIS does not apply a fixed correction to the CNN output; rather, the risk varies according to the prevailing meteorological conditions.

Table 23. Spatial distribution of wildfire risk classes for the evaluated meteorological scenarios.

Scenario	Risk Class	Pixel Count	Coverage (%)	Area (m^2)	Area (km^2)
Scenario 1	Extreme (Red)	45,640	11.14	52,486.0	0.0525
	High (Yellow)	313,679	76.58	360,730.8	0.3607
	Very High (Orange)	20,916	5.11	24,053.4	0.0241
	Total Detected	380,235	92.83	437,270.2	0.4373
Scenario 2	Moderate (Green)	66,556	16.25	76,539.4	0.0766
	Low (Blue)	313,679	76.58	360,730.8	0.3607
	Total Detected	380,235	92.83	437,270.2	0.4373
Scenario 3	Extreme (Red)	66,556	16.25	76,539.4	0.0766
	Very High (Orange)	313,679	76.58	360,730.8	0.3607
	Total Detected	380 235	92.83	437,270.2	0.4373
Scenario 4	Very High (Orange)	45,640	11.14	52,486.0	0.0525
	Moderate (Green)	313,679	76.58	360,730.8	0.3607
	High (Yellow)	20,916	5.11	24,053.4	0.0241
	Total Detected	380,235	92.83	437,270.2	0.4373

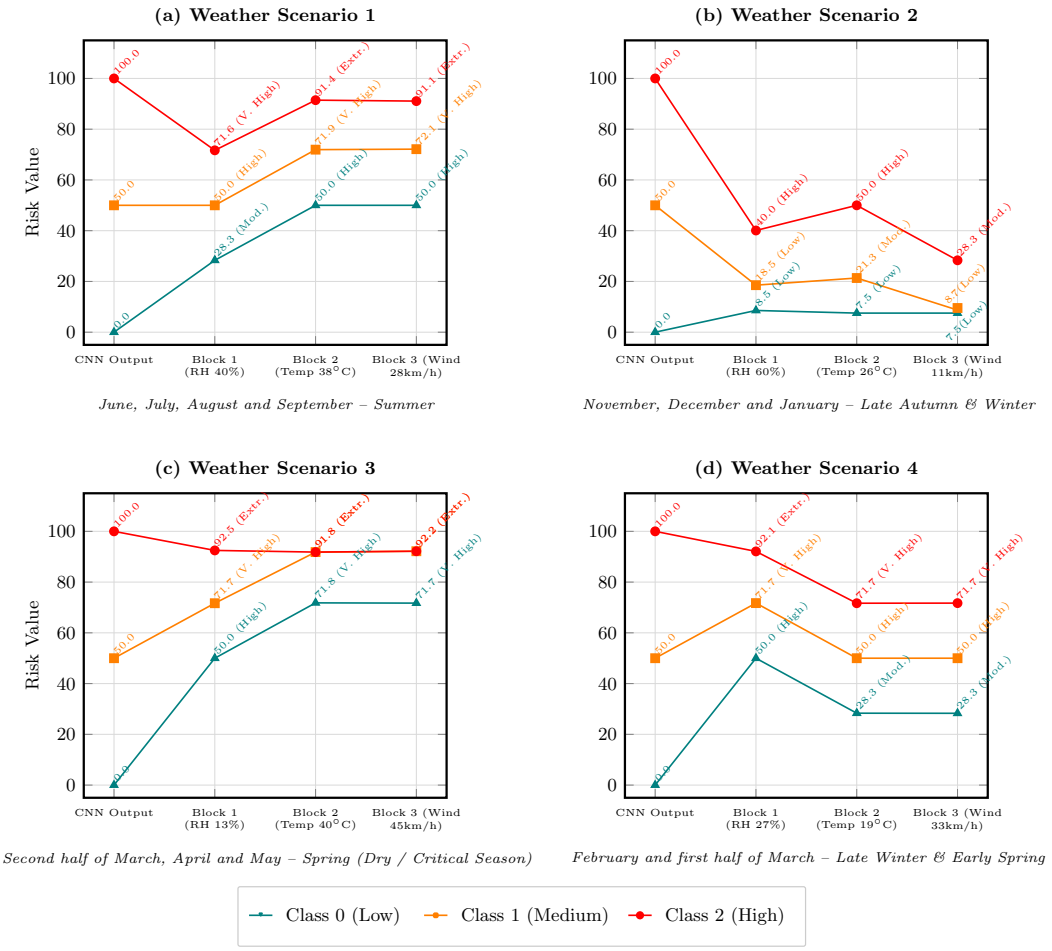


Figure 17. Multi-scenario comparative analysis of the hierarchical dynamics within the Fuzzy Inference System (FIS). The trajectories show how the CNN-derived susceptibility level is modulated under four representative weather scenarios.

Under favorable conditions, as in Weather Scenario 2, the inferred risk decreases, particularly for areas initially classified as highly susceptible. In contrast, under dry and high-temperature conditions, as in Weather Scenario 3, the estimated risk increases across

all susceptibility levels, reflecting greater ignition and propagation potential. Weather Scenarios 1 and 4 show intermediate behavior, indicating that the framework balances the combined effect of humidity, temperature, and wind speed.

4.6. Comparison Between CNN-Based and CNN + FIS

To quantify the specific contribution of the fuzzy inference stage, an additional comparison was performed using the CNN-only output as a baseline. In this baseline, the semantic classes predicted by the U-Net ResNet-50 model were directly mapped into numerical risk values without incorporating meteorological information: low risk was assigned a value of 0, medium risk a value of 50, and high risk a value of 100. These fixed values represent the purely visual risk estimation obtained from the CNN segmentation model. The proposed CNN + FIS framework was then compared against this baseline by evaluating the final fuzzy output under the four meteorological scenarios considered in the case study, as shown in Table 24.

Table 24. CNN-based and CNN + FIS comparison.

Scenario	CNN Class	CNN-Based	CNN + FIS	Variation	Effect
1	Low	0	50.00	+50.00	Increase
1	Medium	50	72.13	+22.13	Increase
1	High	100	91.07	−8.93	Slight decrease
2	Low	0	7.52	+7.52	Stable low risk
2	Medium	50	8.72	−41.28	Decrease
2	High	100	28.33	−71.67	Strong decrease
3	Low	0	71.71	+71.71	Strong increase
3	Medium	50	92.16	+42.16	Strong increase
3	High	100	92.16	−7.84	Extreme maintained
4	Low	0	28.29	+28.29	Moderate increase
4	Medium	50	50.00	0.00	Stable
4	High	100	71.71	−28.29	Decrease

The comparison shows that the CNN-based baseline provides a fixed visual risk classification, whereas the CNN + FIS framework dynamically adjusts the final risk according to humidity, temperature, and wind speed. In low-criticality conditions, such as Scenario 2, the fuzzy inference system reduces the risk initially assigned by the CNN, particularly for the high-risk class, which decreases from 100 to 28.33. Conversely, under critical meteorological conditions, such as Scenario 3, the fuzzy system increases the low and medium CNN classes to substantially higher risk values. Scenario 4 illustrates an intermediate modulation effect, where the low-risk CNN class increases to 28.29, the medium-risk class remains stable at 50.00, and the high-risk class decreases to 71.71. These results indicate that the proposed fuzzy layer does not merely reproduce the CNN output, but acts as an interpretable meteorological modulation stage that adjusts the final risk according to the prevailing environmental conditions.

4.7. Multi-Site Evaluation of the Proposed Framework

To further evaluate the applicability of the proposed framework beyond the illustrative case presented in the previous section, three additional georeferenced RGB images acquired from different locations in southern Mexico were selected for analysis, summarized in Tables 25–27. These study sites were chosen because they represent distinct environmental settings within one of the regions of the country that historically reports the highest annual wildfire occurrence. For each location, the corresponding geographic coordinates and representative meteorological conditions are provided to facilitate the interpretation of

the obtained wildfire risk maps. The spatial distribution of the estimated wildfire risk for Study Site 1 is shown in Figure 18. The figure compares the original aerial image with the initial CNN segmentation and the successive outputs of Blocks 1–3, illustrating how the hierarchical fuzzy inference system progressively adjusts the risk map according to the meteorological conditions. In this case, the initially detected high-risk regions evolve toward very high and extreme risk levels after incorporating humidity, temperature, and wind speed.

Table 25. Risk assessment results for study site 1.

Coordinates	**Latitude: North: 16.3948180°, South: 16.3821739° Longitude: West: −91.214190°, East: −97.106456°**		
Conditions	**Humidity = 40%, Temperature = 38 °C, Wind = 28 km/h**		
CNN	**Block 1**	**Block 2**	**Block 3**
0	28.28 (Moderate)	50 (High)	50 (High)
1	50 (High)	71.75 (Very high)	71.81 (Very high)
2	71.72 (Very high)	91.44 (Extreme)	91.82 (Extreme)

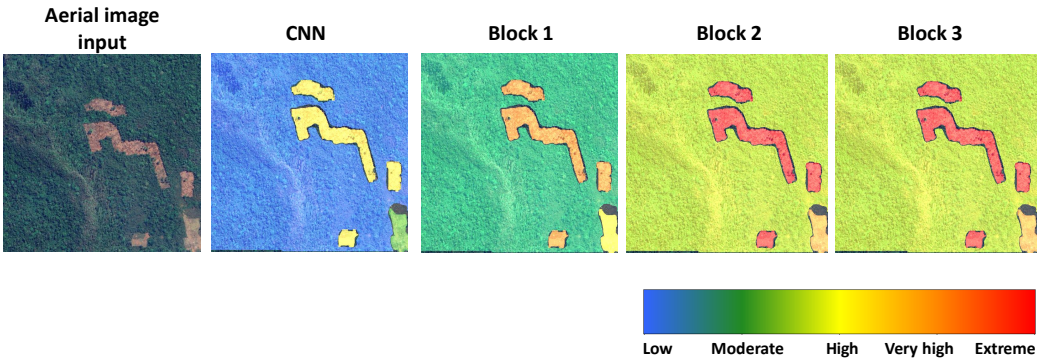


Figure 18. Risk segmentation results for Study site 1.

The corresponding wildfire risk segmentation for Study Site 2 is presented in Figure 19. The figure compares the original aerial image with the CNN-based classification and the successive outputs of the three fuzzy inference blocks. The regions initially identified as moderate risk remain largely unchanged after incorporating humidity in Block 1, but are subsequently reclassified as high risk after accounting for temperature and wind speed in Blocks 2 and 3.

The resulting wildfire risk segmentation for Study Site 3 is illustrated in Figure 20. The figure compares the original aerial image with the CNN classification and the successive outputs of the hierarchical fuzzy inference blocks. The CNN initially identifies most of the scene as high risk, with a localized low-risk region. After incorporating humidity, temperature, and wind speed, the dominant areas are progressively shifted toward extreme risk, while the initially low-risk region increases to high and very high levels.

Table 26. Risk assessment results for study site 2.

Coordinates	**Latitude—North: 16.4349665°, South: 16.4260432° Longitude—West: −92.3406189°, East: −92.3156064°**		
Conditions	**Humidity = 48%, Temperature = 27 °C, Wind = 31 km/h**		
CNN	**Block 1**	**Block 2**	**Block 3**
0	17.26 (Low)	16.40 (Low)	13.35 (Low)
1	38.04 (Moderate)	40.17 (High)	50 (High)

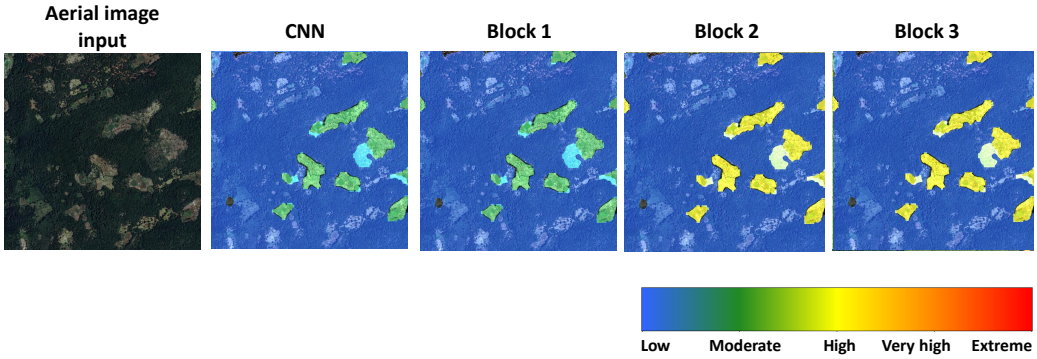


Figure 19. Risk segmentation results for Study site 2.

Table 27. Risk assessment results for study site 3.

Coordinates	**Latitude—North: 18.5794589°, South: 18.5729431° Longitude—West: −90.3725885°, East: −90.3543244°**		
Conditions	**Humidity = 20%, Temperature = 34 °C, Wind = 29 km/h**		
CNN	**Block 1**	**Block 2**	**Block 3**
0	50 (High)	72.13 (Very high)	72.31 (Very high)
2	91.44 (Extreme)	91.07 (Extreme)	91.44 (Extreme)

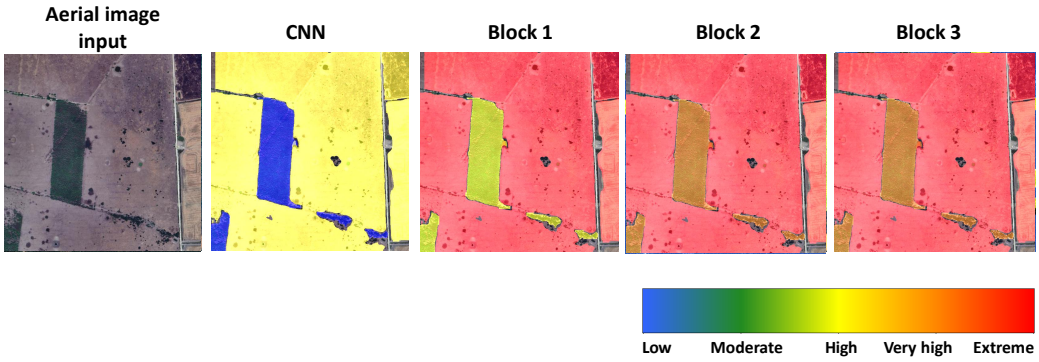


Figure 20. Risk segmentation results for Study site 3.

Unlike the previous analysis, which focused on evaluating the dynamic response of the hierarchical fuzzy inference system under representative seasonal meteorological scenarios, the objective of this section is to examine the behavior of the complete CNN + FIS framework across different landscapes and environmental contexts. The selected study sites exhibit variations in topography, vegetation cover, land use, and climatic conditions, providing a broader evaluation of the proposed methodology under representative wildfire-prone environments.

5. Discussion

The results obtained in this study demonstrate the effectiveness of combining deep learning and fuzzy logic for wildfire risk estimation. Experimental evaluations under different meteorological scenarios showed that the proposed neuro-fuzzy system dynamically modulates the initial CNN-based segmentation according to environmental conditions associated with wildfire ignition and propagation. In particular, the system was capable of attenuating or increasing the estimated risk depending on variations in relative humidity, temperature, and wind speed, generating spatial risk distributions that are more consistent with expected wildfire behavior than those obtained from the CNN classification alone. The comparative evaluation of semantic segmentation models showed that the U-Net

architecture with a ResNet50 backbone achieved the best performance across all evaluation metrics, particularly in terms of F1-Score and Intersection over Union (IoU). This indicates a superior capability to accurately delineate spatial patterns associated with fire-prone areas, such as vegetation distribution and terrain characteristics.

The integration of the CNN output into the fuzzy inference system proved to be a key component of the proposed methodology. By using the categorical output of the segmentation model as an input variable, the system is able to incorporate high-level visual information into the decision-making process. This approach allows the initial risk estimation derived from image analysis to be dynamically adjusted according to meteorological conditions, resulting in a more context-aware and realistic assessment.

The hierarchical structure of the fuzzy inference system plays a crucial role in modeling the different stages of wildfire dynamics. In the proposed architecture, the initial inference stage prioritizes vegetation-related information obtained from the CNN together with relative humidity, reflecting their strong influence on fuel condition and ignition susceptibility. Subsequent stages incorporate temperature and wind speed to model the intensification and propagation behavior of wildfires under adverse environmental conditions.

The distinct behaviors observed across the weather scenarios validate the biophysical prioritization embedded in this hierarchical structure. By processing relative humidity in Block 1 immediately after the CNN spatial classification, the system mimics real-world fuel moisture dynamics. For instance, in Scenario 3, the extreme 13% humidity deficit produced an upward shift in the baseline risk, demonstrating that regardless of wind or temperature, fuel dryness is the primary gatekeeper for ignition. Conversely, as seen in Scenario 1 and Scenario 2, when relative humidity increased to 40% and 60% respectively, it acted as an immediate environmental buffer. This early filtering prevents the system from triggering over-saturated risk maps, a common failure in linear weighting models where a high temperature alone can cause false alarms in high-moisture environments.

Experimental results demonstrated that the hierarchical structure progressively modulates the wildfire risk estimation according to the analyzed meteorological variables, producing coherent transitions between risk levels across the evaluated scenarios. Furthermore, the hierarchical decomposition significantly reduced the complexity of the fuzzy rule base. While a conventional fuzzy system combining all variables simultaneously would require approximately 375 rules, the proposed hierarchical approach reduced the system to approximately 65 rules distributed across independent inference stages. This reduction improves interpretability and computational efficiency while preserving the capability to model nonlinear wildfire behavior.

This reduction from 375 to approximately 65 fuzzy rules does not merely represent a computational advantage; it directly enhances spatial and causal interpretability. Because the rules are segmented into independent, thematic inference stages, stakeholders can trace exactly why a specific pixel transitioned from 'Low' risk in the CNN stage to 'Extreme' in the final output. It is possible to determine whether a high-risk zone is mainly driven by severe fuel dryness (Block 1), critical thermal load (Block 2), or high propagation velocity due to wind (Block 3). This level of transparency bridges the gap between complex deep learning models and actionable decision-making in the field.

Furthermore, Scenario 4 highlights the system's capability to resolve conflicting environmental inputs, an essential requirement for operational risk management. This specific case simulated a high-wind, low-moisture, but low-temperature environment (typical of seasonal dry fronts). A conventional multi-criteria evaluation might average these values out, resulting in a misleading 'moderate' risk across the board. In contrast, the proposed FIS successfully modeled a 'tug-of-war' behavior: Block 1 detected the high ignition susceptibility due to the 27% humidity deficit, Block 2 applied a thermal brake due to the cool 19 °C

temperature, and Block 3 factored in the 33 km/h wind as a severe propagation threat. The resulting final output ('Very High' risk in vulnerable areas) proves that the system preserves critical threat indicators even when some atmospheric variables appear favorable.

Another important advantage of the proposed approach is its interpretability. Unlike purely data-driven models, the fuzzy inference system provides a transparent reasoning process through linguistic rules and membership functions. This makes the model more suitable for real-world applications, where understanding the influence of different variables is essential for decision-making and risk management.

It is also important to clarify that the proposed methodology is not intrinsically restricted to a single geographical region. The framework is based on general input sources, namely georeferenced RGB aerial imagery, meteorological variables, semantic segmentation, and fuzzy inference. Therefore, the same methodological structure can be transferred to other regions, provided that representative imagery, local meteorological records, and region-specific annotation criteria are available. In this sense, the main adaptation required for new geographic contexts would involve retraining or fine-tuning the segmentation model with local image data and adjusting the fuzzy membership functions according to regional wildfire risk thresholds and expert knowledge.

Although the dataset used in this study is regional, it should not be interpreted as a single-site dataset. The imagery covers a broad territory within southern Mexico and includes different locations, vegetation conditions, terrain structures, and meteorological contexts. This regional variability allowed the proposed framework to be evaluated under heterogeneous environmental conditions rather than under a single isolated scene.

The modular design of the framework also provides flexibility for deployment in other regions. However, successful implementation in new environments would require representative imagery, calibration of meteorological input ranges, and incorporation of local expert knowledge to ensure consistency with regional wildfire behavior. A full cross-regional validation based on independent ecological regions, historical wildfire inventories, and burned-area records remains necessary and will be addressed in future work.

Despite these promising results, some limitations should be considered. The dataset used in this study is relatively limited in size, which may affect the generalization capability of the trained models. Additionally, the use of a fixed set of meteorological variables may not fully capture the complexity of wildfire behavior, as other factors such as topography, fuel type, and temporal dynamics also play a significant role. Future work should focus on expanding the dataset, incorporating additional environmental variables, and exploring spatiotemporal modeling approaches to further enhance the accuracy and robustness of the system. Although the dataset includes high-resolution georeferenced imagery from different locations in southern Mexico, its size and seasonal coverage remain limited. Future work will expand the dataset by incorporating additional images from different seasons, broader topographic gradients, and ecological regions, as well as external validation using official wildfire inventories or burned-area records.

The meteorological scenarios presented in this study should be interpreted as controlled sensitivity analyses rather than as a complete operational validation of wildfire occurrence. Their purpose is to evaluate how the proposed fuzzy inference system modulates the initial CNN-based risk classification under different humidity, temperature, and wind-speed conditions. Although this analysis demonstrates the internal consistency and interpretability of the proposed framework, external validation using historical wildfire occurrence locations, burn scar data, or official wildfire inventories remains necessary to quantify predictive accuracy under real operational conditions. This aspect will be addressed in future work.

The reported segmentation metrics (IoU, Accuracy, Precision, Recall, and F1-score) quantify the agreement between the predicted semantic masks and the manually annotated reference labels. Therefore, these metrics evaluate the ability of the proposed CNN to reproduce the predefined wildfire susceptibility classes from high-resolution imagery. Complementary validation of the final neuro-fuzzy risk maps using historical wildfire occurrence data or operational fire-danger indices would further extend the assessment of the proposed framework in future studies.

6. Conclusions

This study presented a hybrid approach for forest fire risk estimation that integrates a semantic segmentation CNN with a hierarchical fuzzy inference system. The proposed methodology effectively combines visual information extracted from high-resolution georeferenced RGB imagery with meteorological parameters to produce a comprehensive and interpretable risk assessment.

The results demonstrated that the ResNet50-based U-Net model provides the most accurate segmentation performance, serving as a reliable input for the fuzzy inference system. The additional comparison with Mask2Former, OneFormer, and Segment Anything confirmed the competitive advantage of U-Net ResNet-50, which achieved the best overall results among the evaluated models. These findings support its selection as the segmentation module of the proposed neuro-fuzzy framework.

Furthermore, the hierarchical FIS successfully models the influence of environmental factors on both ignition and propagation processes. This behavior was evidenced through the experimental scenarios, where variations in relative humidity, temperature, and wind speed produced coherent modifications in the estimated wildfire risk maps. In particular, low humidity and high temperature conditions increased the extent of high-risk regions, while stronger wind conditions intensified the propagation-related risk levels in subsequent inference stages. These results demonstrate that the proposed system generates risk estimations consistent with the expected environmental dynamics of wildfire behavior.

The evaluation of specific case studies further illustrated the dynamic sensitivity and robustness of the hierarchical FIS structure. Under extreme conditions mimicking the end of the dry season—characterized by intense thermal loads, critical humidity deficits, and high wind speeds—the system effectively simulated high-danger scenarios by escalating risks to maximum categories without over-saturation. Conversely, during low-criticality or transitional scenarios, the system proved its capability to downscale or stabilize risk indicators, highlighting the role of relative humidity as an immediate environmental buffer in the initial stage. Furthermore, when tested against conflicting variables, such as low temperatures paired with high wind speeds and dry air, the model demonstrated an consistent ability to balance opposing forces, preventing false alarms while accurately maintaining appropriate vigilance levels. This granular analysis confirms that the proposed framework does not rely on a single dominant variable, but rather synthesizes multi-environmental factors to reflect real-world wildfire vulnerabilities.

The main contribution of this work lies in the integration of deep learning and expert knowledge through a hierarchical fuzzy framework, enabling a balance between predictive performance and interpretability. This approach represents a valuable tool for supporting wildfire monitoring and decision-making processes, since the proposed neuro-fuzzy system generates spatially interpretable wildfire risk maps that dynamically adapt to changing environmental conditions. The experimental results demonstrated that the integration of meteorological variables modifies the spatial distribution and extent of the detected risk regions, allowing the identification of areas with increased ignition susceptibility and propagation potential. Furthermore, the estimation of physical coverage areas associated

with each risk level provides quantitative information that can support environmental analysis, risk prioritization, and preventive monitoring strategies.

Future work will focus on incorporating additional environmental variables, expanding the dataset, and exploring temporal analysis to further improve the performance and applicability of the proposed system.

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